

Probability 1 Notes

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1 Games of Chance (A.N. Kolmogorov)

Random Experiment: A random experiment is a procedure in which repeated trials may result in different outcomes.

Uncertainty: The purpose of probability theory is to quantify uncertainty.

1.1 Sample Space

Definition: The sample space of a random experiment, denoted by Ω , is the collection of all possible outcomes.

$$\Omega = \{\text{all possible outcomes}\}$$

Examples:

$$\text{Toss of a die: } \Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\text{Toss of a coin: } \Omega = \{H, T\}$$

Note: Infinity is not an element of Ω , i.e. $\infty \notin \Omega$, but Ω may contain infinitely many elements.

Example of Sample Space

Time it takes for you to come from residence to class daily:

$$\Omega = [0, 15]$$

Exercise 1

Identify 5 different random experiments and write down their sample spaces.

1.2 Events

An **event** is any subset of the sample space:

$$A \subseteq \Omega$$

1.3 Digression

Example:

$$A = \{1\}, \quad A^c = \{2, 3, 4, 5\}, \quad A \subseteq \mathbb{N}$$

2 Mutually Exclusive Events

Consider

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \quad A = \{2, 4\}, \quad B = \{1, 3, 5\}$$

If the outcome is 4, then event A has occurred while event B has not.

2.1 Set Theory Review

$$U = \{1, 2, 3, 4, 5, 6\}, \quad A = \{2, 4\}, \quad B = \{1, 3, 5\}$$

$$A^c = U - A = \{1, 3, 5\}$$

Disjoint Events: Two sets are disjoint if

$$A \cap B = \emptyset$$

Properties of Sets

- Reflexive: $A \subseteq A$
- Transitive: $A \subseteq B, B \subseteq C \Rightarrow A \subseteq C$
- Symmetric (Equality): $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$
- Complementarity:

$$A \subseteq B \Rightarrow B^c \subseteq A^c$$

Partition Property

A and A^c form a partition of U :

$$A \cup A^c = U, \quad A \cap A^c = \emptyset$$

Terminology: Event A occurs if the outcome ω lies in A .

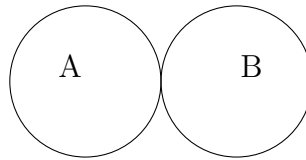
3 Basic Set Operations

$$A \subseteq B \iff (\omega \in A \Rightarrow \omega \in B)$$

$$A^c = \Omega - A = \Omega \cap A^c$$

$$A \cup B = \{\omega : \omega \in A \text{ or } \omega \in B\}$$

$$A \cap B = \{\omega : \omega \in A \text{ and } \omega \in B\}$$



Venn diagram for mutually exclusive events: $A \cap B = \emptyset$.

4 Probability

For every event A , $P(A)$ quantifies uncertainty.

4.1 Axioms of Probability

1. $P(A) \geq 0$ for any event A .
2. $P(\Omega) = 1$.
3. (Countable additivity) If $A_1, A_2, \dots$ are mutually exclusive,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

4.2 Countable Sets

A set E is countable if there exists a one-to-one correspondence between E and $\mathbb{N}$.

4.3 Probability Measure

Any function P satisfying axioms (1), (2), (3) is called a **probability measure**. The triple $(\Omega, \mathcal{F}, P)$ is called a **probability space**.

5 Properties of Probability

5.1 1. $P(\emptyset) = 0$

Proof: Let $A_i = \emptyset$ for all i . Then

$$\bigcup_{i=1}^{\infty} A_i = \emptyset$$

By countable additivity,

$$P(\emptyset) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Let $P(\emptyset) = p$. Then

$$p = p + p + p + \dots$$

If $p > 0$, the RHS diverges to ∞ , contradiction. Hence,

$$P(\emptyset) = 0.$$

5.2 2. Finite Additivity

If $A_1, A_2, \dots, A_n$ are mutually exclusive:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

6 Further Properties of Probability

6.1 Finite Additivity (Completed Proof)

Define

$$A_{n+1} = A_{n+2} = \dots = \emptyset.$$

Then the sequence $A_1, A_2, \dots$ becomes mutually exclusive.

Also,

$$\bigcup_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^n A_i \right) \cup \left(\bigcup_{i=n+1}^{\infty} \emptyset \right) = \bigcup_{i=1}^n A_i.$$

By countable additivity,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^n P(A_i) + \sum_{i=n+1}^{\infty} P(\emptyset) = \sum_{i=1}^n P(A_i).$$

Thus the finite additivity formula holds:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

6.2 Example

In an MCQ, the probability of choosing the correct option is 0.2. Let A = “correct option is chosen”. Then

$$A^c = \text{“incorrect option chosen”}, \quad P(A) = 0.2.$$

Since $A \cup A^c = \Omega$ and $A \cap A^c = \emptyset$,

$$1 = P(\Omega) = P(A) + P(A^c) \Rightarrow P(A^c) = 0.8.$$

6.3 Property 3: Complement Rule

For any event A ,

$$P(A^c) = 1 - P(A).$$

6.4 Property 4: Bounds on Probability

For any event A ,

$$0 \leq P(A) \leq 1.$$

6.5 Example: Two Components in Series

Two identical components connected in series. Probability each component fails: 0.1. So probability it works: 0.9.

Let

$$F_j = \text{component } j \text{ fails,} \quad S_j = \text{component } j \text{ works.}$$

System works only if both work:

$$P(S_1 \cap S_2) = 0.9 \times 0.9 = 0.81.$$

Thus system fails:

$$P(A) = 1 - 0.81 = 0.19.$$

6.6 Property 5: Two-set Formula

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

6.7 Example

Given:

$$P(A) = 0.6, \quad P(B) = 0.8, \quad P(A \cap B) = 0.5.$$

(1) Probability household has at least one connection:

$$P(A \cup B) = 0.9.$$

(2) Household has neither connection:

$$P((A \cup B)^c) = 0.1.$$

(3) Household has exactly one:

$$P((A \setminus B) \cup (B \setminus A)) = P(A \cup B) - P(A \cap B) = 0.4.$$

6.8 Property 6: Three-set Inclusion–Exclusion

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(B \cap C) - P(C \cap A) \\ &\quad + P(A \cap B \cap C). \end{aligned}$$

6.9 Property 7: General Inclusion–Exclusion

For events $A_1, \dots, A_n$,

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) \\ &\quad - \dots + (-1)^{n+1} P\left(\bigcap_{i=1}^n A_i\right). \end{aligned}$$

Proof (Completed)

Base case: For $n = 2$,

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).$$

Inductive step: Assume the result for $n = k$. Then

$$\bigcup_{i=1}^{k+1} A_i = \left(\bigcup_{i=1}^k A_i \right) \cup A_{k+1}.$$

Apply two-set formula:

$$P\left(\bigcup_{i=1}^{k+1} A_i\right) = P\left(\bigcup_{i=1}^k A_i\right) + P(A_{k+1}) - P\left(\left(\bigcup_{i=1}^k A_i\right) \cap A_{k+1}\right).$$

Also,

$$\left(\bigcup_{i=1}^k A_i\right) \cap A_{k+1} = \bigcup_{i=1}^k (A_i \cap A_{k+1}).$$

Now apply the induction hypothesis to that union and simplify. Result follows.

6.10 Property 8: Union Bound

$$P(A \cup B) \leq P(A) + P(B).$$

6.11 Property 9: Finite Subadditivity

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

6.12 Property 10: Countable Subadditivity

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i).$$

6.13 Property 11: Continuity from Below

If

$$A_1 \subseteq A_2 \subseteq \cdots,$$

then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P(A_n).$$

Continuation of Property 11 (Continuity from Below)

Define the sets

$$B_1 = A_1, \quad B_2 = A_2 \cap A_1^c, \quad B_3 = A_3 \cap A_2^c, \quad \dots$$

and in general

$$B_{n+1} = A_{n+1} \cap A_n^c.$$

Then the sets $B_1, B_2, \dots$ are mutually exclusive and

$$\bigcup_{i=1}^n B_i = A_n.$$

Thus,

$$P(A_n) = P\left(\bigcup_{i=1}^n B_i\right) = \sum_{i=1}^n P(B_i).$$

Now,

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i.$$

Hence,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} P(B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(B_i) = \lim_{n \rightarrow \infty} P(A_n).$$

Note: The sequence $\{P(A_1), P(A_2), \dots\}$ is increasing because the events are increasing. Thus $\lim_{n \rightarrow \infty} P(A_n)$ exists and equals

$$P\left(\bigcup_{i=1}^{\infty} A_i\right).$$

Let $A = \bigcup_{i=1}^{\infty} A_i$. Then

$$P(A_n) \uparrow P(A).$$

6.14 Property 12: Continuity from Above

Let

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$$

be a decreasing sequence of events and let

$$A = \bigcap_{i=1}^{\infty} A_i.$$

Then

$$P(A_n) \downarrow P(A), \quad \text{i.e.} \quad P(A) = \lim_{n \rightarrow \infty} P(A_n).$$

Proof: Define $B_n = A_1 \setminus A_n$. Then the sequence B_n is increasing, because

$$A_1 \setminus A_{n+1} \supseteq A_1 \setminus A_n.$$

Also,

$$A_n = A_1 \setminus B_n.$$

Applying continuity from below to B_n :

$$P(B_n) \uparrow P\left(\bigcup_{i=1}^{\infty} B_i\right) = P(A_1 \setminus A).$$

Hence,

$$P(A_n) = P(A_1) - P(B_n) \downarrow P(A_1) - P(A_1 \setminus A) = P(A).$$

7 Assigning Probability Systematically

Suppose Ω is countable. Then either

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$$

(finite) or

$$\Omega = \{\omega_1, \omega_2, \omega_3, \dots\}$$

(countably infinite).

Let E_i be simple events where $E_i = \{\omega_i\}$. Assign $P(E_i) \geq 0$ such that

$$\sum_{i=1}^{\infty} P(E_i) = 1.$$

For any event $A \subseteq \Omega$,

$$P(A) = \sum_{\omega_i \in A} P(E_i).$$

Example: Repeated coin toss until first head

$$\Omega = \{H, TH, TTH, TTTH, \dots\}.$$

Let $P(H) = p$, $P(T) = 1 - p$, $0 \leq p \leq 1$.

Then:

$$P(TTH) = (1 - p)^2 p, \quad P(\underbrace{TT \cdots T}_{n \text{ times}} H) = (1 - p)^n p.$$

Special case: fair coin $p = \frac{1}{2}$.

Check:

$$\sum_{n=0}^{\infty} p(1 - p)^n = \frac{p}{1 - (1 - p)} = 1.$$

Valid for $0 < p < 1$.

Example

Let $\Omega = \{1, 2, 3, 4, 5\}$.

Let $P(i) = \frac{i}{10}$. Then

$$P(\{2, 3\}) = P(2) + P(3) = \frac{2+3}{10} = \frac{1}{2}.$$

Exercise: Toss a coin until first head

Let A = “number of tosses is even.”

Outcome sample space:

$$\Omega = \{TH, TTTH, TTTTTH, \dots\}$$

For event A :

$$P(TH) = (1-p)p, \quad P(TTTH) = (1-p)^3p, \quad \dots$$

Thus:

$$P(A) = \sum_{n=0}^{\infty} (1-p)^{2n+1}p = p(1-p) \sum_{n=0}^{\infty} (1-p)^{2n} = \frac{p(1-p)}{1-(1-p)^2} = \frac{1-p}{2-p}.$$

8 Equally Likely Model

Suppose

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$$

and all outcomes are equally likely.

Then

$$P(\{\omega_i\}) = \frac{1}{N}, \quad 1 \leq i \leq N.$$

Thus,

$$P(A) = \sum_{\omega_i \in A} \frac{1}{N} = \frac{|A|}{|\Omega|}.$$

9 Counting Techniques

9.1 Equally Likely Outcomes

If Ω is a finite sample space with $|\Omega| = N$ and all outcomes are equally likely, then

$$P(\{\omega\}) = \frac{1}{N}, \quad \omega \in \Omega$$

and for any event $A \subseteq \Omega$,

$$P(A) = \frac{|A|}{|\Omega|}.$$

9.2 Combinations

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad 0 \leq k \leq n.$$

This is the number of ways of selecting k objects from n objects when order does not matter.

9.3 Permutations

$${}^nP_k = \frac{n!}{(n-k)!}.$$

This is the number of ordered selections of k objects from n .

9.4 Example

A team of 15 players contains:

7 batsmen, 6 bowlers, 2 all-rounders.

Number of ways to form a team of: - 5 batsmen, - 5 bowlers, - 1 all-rounder:

$$\binom{7}{5} \cdot \binom{6}{5} \cdot \binom{2}{1} = 21 \cdot 6 \cdot 2 = 252.$$

If batting order is required, then each of the 11 players may appear in any order:

$252 \times 11!$ possible batting orders.

9.5 Example: Tossing a Fair Die

Let $A = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$ be the outcomes where the sum is 5.

$$P(A) = \frac{4}{36} = \frac{1}{9}.$$

9.6 Example: Tossing 3 Fair Coins

Ω has 8 outcomes.

Let $A =$ “at least two heads”:

$$A = \{HHH, HHT, HTH, THH\}, \quad |A| = 4.$$

$$P(A) = \frac{4}{8} = \frac{1}{2}.$$

Let $B =$ “first toss is head”. Then $P(B) = 4/8 = 1/2$.

Given B occurred, the favorable outcomes for A are:

$$\{HHH, HHT, HTH\}.$$

Thus,

$$P(A \mid B) = \frac{3}{4}.$$

10 Conditional Probability

For events A and B with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Useful identity:

$$P(A^c \cap B) = P(B) - P(A \cap B).$$

Another:

$$P(A^c \mid B) = 1 - P(A \mid B).$$

11 Example: Defective Components Problem

A factory has 2 assembly lines:

Line 1: 10 components, 2 defective, 8 good

Line 2: 15 components, 1 defective, 14 good

Let A_1 = component is from line 1, A_2 = component is from line 2.

$$P(A_1) = \frac{10}{25} = 0.4, \quad P(A_2) = \frac{15}{25} = 0.6.$$

Let D = component is defective.

$$P(D \mid A_1) = \frac{2}{10} = 0.2, \quad P(D \mid A_2) = \frac{1}{15}.$$

Using total probability:

$$\begin{aligned} P(D) &= P(D \mid A_1)P(A_1) + P(D \mid A_2)P(A_2) \\ &= 0.2 \cdot 0.4 + \frac{1}{15} \cdot 0.6 = \frac{3}{25}. \end{aligned}$$

Given that the component is defective, probability it came from line 1:

$$P(A_1 \mid D) = \frac{P(D \mid A_1)P(A_1)}{P(D)} = \frac{2/25}{3/25} = \frac{2}{3}.$$

Exercise

Let $P(B) > 0$. Define a new function:

$$Q(A) = P(A \mid B), \quad A \subseteq \Omega.$$

Show that $Q(\cdot)$ is a probability measure on Ω .

12 Multiplication Rule

For any two events A and B with $P(B) > 0$,

$$P(A \cap B) = P(B) P(A \mid B).$$

If $A_1, A_2, \dots, A_n$ are mutually exclusive and exhaustive, then for any event B with $P(B) > 0$,

$$P(B) = \sum_{i=1}^n P(A_i) P(B \mid A_i).$$

This is the **Law of Total Probability**.

12.1 Example: Drawing Balls Without Replacement

A box contains 8 balls: 5 orange and 3 green. Two balls are drawn one after another *without* replacement.

Let

$$O_i = \{i\text{-th ball is orange}\}, \quad G_i = \{i\text{-th ball is green}\}, \quad i = 1, 2.$$

We want the probability that the second ball is green:

$$P(G_2).$$

Use the total probability decomposition:

$$G_2 = (G_2 \cap G_1^c) \cup (G_2 \cap G_1).$$

Compute:

$$\begin{aligned} P(G_2 \mid G_1) &= \frac{2}{7}, & P(G_1) &= \frac{3}{8}, \\ P(G_2 \mid O_1) &= \frac{3}{7}, & P(O_1) &= \frac{5}{8}. \end{aligned}$$

Thus,

$$P(G_2) = P(G_2 \mid G_1)P(G_1) + P(G_2 \mid O_1)P(O_1) = \frac{2}{7} \cdot \frac{3}{8} + \frac{3}{7} \cdot \frac{5}{8} = \frac{21}{56}.$$

Exercise

Find:

$$P(G_1 \mid O_2).$$

Using Bayes decomposition:

$$O_2 = (O_1^c \cap O_2) \cup (O_1 \cap O_2),$$

compute both terms:

$$P(O_2) = \frac{5}{8} \cdot \frac{4}{7} + \frac{3}{8} \cdot \frac{5}{7} = \frac{5}{14} + \frac{15}{56} = \frac{35}{56}.$$

Similarly,

$$P(G_1 \cap O_2) = P(G_1)P(O_2 \mid G_1) = \frac{3}{8} \cdot \frac{5}{7} = \frac{15}{56}.$$

Hence,

$$P(G_1 \mid O_2) = \frac{15/56}{35/56} = \frac{3}{7}.$$

13 Law of Total Probability and Bayes' Theorem

Let $Q(A) = P(A \mid B)$ for fixed B with $P(B) > 0$. Then $Q(\cdot)$ is a probability measure on Ω .

13.1 Mutually Exclusive and Exhaustive Events

Events $A_1, A_2, \dots, A_n$ are:

$$A_i \cap A_j = \emptyset \ (i \neq j), \quad \bigcup_{i=1}^n A_i = \Omega.$$

13.2 Law of Total Probability

If $A_1, \dots, A_n$ are mutually exclusive and exhaustive and $P(A_i) > 0$, then for any event B ,

$$P(B) = \sum_{i=1}^n P(A_i)P(B \mid A_i).$$

13.3 Bayes' Theorem

Let $A_1, \dots, A_n$ be mutually exclusive and exhaustive, with $P(A_i) > 0$.

For any event B with $P(B) > 0$,

$$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{\sum_{j=1}^n P(B \mid A_j)P(A_j)}.$$

Useful form:

$$P(A_i \mid B) = \frac{P(A_i \cap B)}{P(B)}.$$

14 Example: Mobile Phone Brands

Three brands of phones are sold:

$$P(A_1) = 0.5, \quad P(A_2) = 0.3, \quad P(A_3) = 0.2.$$

During warranty:

$$P(R \mid A_1) = 0.25, \quad P(R \mid A_2) = 0.20, \quad P(R \mid A_3) = 0.10.$$

Probability a randomly chosen phone needs repair:

$$P(R) = 0.25(0.5) + 0.20(0.3) + 0.10(0.2) = 0.205.$$

Posterior probabilities:

$$\begin{aligned}P(A_1 | R) &= \frac{0.25(0.5)}{0.205} = \frac{0.125}{0.205} \approx 0.61, \\P(A_2 | R) &= \frac{0.20(0.3)}{0.205} = 0.060/0.205 \approx 0.29, \\P(A_3 | R) &= \frac{0.10(0.2)}{0.205} = 0.020/0.205 \approx 0.10.\end{aligned}$$

15 Example: Rare Disease Testing

Disease incidence:

$$P(A_1) = 0.001, \quad P(A_2) = 0.999.$$

Test characteristics:

$$\begin{aligned}P(B | A_1) &= 0.99, & (\text{true positive}) \\P(B | A_2) &= 0.02, & (\text{false positive}).\end{aligned}$$

Probability test is positive:

$$P(B) = 0.99(0.001) + 0.02(0.999) = 0.02097.$$

Probability individual actually has disease:

$$P(A_1 | B) = \frac{0.99(0.001)}{0.02097} \approx 0.047.$$

So even with a **99% accurate test**, the posterior probability is only **4.7

16 Independence of Events

Events A and B are independent if:

$$P(A | B) = P(A) \iff P(A \cap B) = P(A)P(B).$$

16.1 Example: Toss of a Die

Let $E_i = \{i\}$ with $P(E_i) = 1/6$.

Define:

$$A = \{1, 2, 3\}, \quad B = \{2, 4, 6\}, \quad C = \{2, 3, 4, 5\}.$$

Compute:

$$P(A) = \frac{3}{6} = \frac{1}{2}, \quad P(B) = \frac{3}{6} = \frac{1}{2}, \quad P(C) = \frac{4}{6} = \frac{2}{3}.$$

Check independence:

$$A \cap B = \{2\} \Rightarrow P(A \cap B) = \frac{1}{6},$$

$$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \neq \frac{1}{6}.$$

Thus A and B are not independent.

Similarly check B and C :

$$B \cap C = \{2, 4\} \Rightarrow P(B \cap C) = \frac{2}{6} = \frac{1}{3},$$

$$P(B)P(C) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

So B and C are independent.

16.2 Further Results on Independence

Proof

Suppose

$$P(A \mid B) = P(A).$$

Then

$$P(A \cap B) = P(A \mid B)P(B) = P(A)P(B).$$

Hence A and B are independent.

16.3 Definition

Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

16.4 Example

If A and B are mutually exclusive and $P(A) > 0$, then

$$P(A \cap B) = 0 \neq P(A)P(B),$$

so A and B cannot be independent.

16.5 Example: Can A be independent of itself?

Yes, only if

$$P(A) = 0 \quad \text{or} \quad P(A) = 1.$$

16.6 Equivalent Condition

A and B are independent if and only if

$$P(A \mid B) = P(A) \quad \text{and therefore} \quad P(B \mid A) = P(B).$$

16.7 More Properties

If A and B are independent, then so are

$$A \cap B^c, \quad A^c \cap B, \quad A^c \cap B^c.$$

16.8 Trivial Case

If $P(A) = 0$ or $P(A) = 1$, then A is independent of *every* event.

16.9 Mutual Independence

Let $A_1, \dots, A_n$ be events. They are said to be **mutually independent** if for every $2 \leq k \leq n$ and every set of indices

$$1 \leq i_1 < i_2 < \dots < i_k \leq n,$$

we have

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \prod_{j=1}^k P(A_{i_j}).$$

16.10 Example (Pairwise vs Mutual Independence)

For $n = 3$, pairwise independence requires:

$$P(A_1 \cap A_2) = P(A_1)P(A_2),$$

$$P(A_1 \cap A_3) = P(A_1)P(A_3),$$

$$P(A_2 \cap A_3) = P(A_2)P(A_3).$$

Mutual independence additionally requires:

$$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3).$$

16.11 Bernoulli Trials and Repeated Experiments

Consider an experiment with two possible outcomes: Success S and Failure F with

$$P(S) = p, \quad 0 < p < 1.$$

Repeated (independent) trials yield outcomes such as

$$\{F, F, \dots, F, S\}.$$

Then

$$P(\underbrace{FF \dots FS}_{n\text{th trial} = S}) = (1 - p)^{n-1}p.$$

16.12 Example: Circuit System

Three components arranged as shown:

(Series of 1, then parallel of 2 and 3)

All components work independently with probability p . Let A_i = event that component i works.

System fails only if: - Component 1 fails, or - Both components 2 and 3 fail.

Thus system works if:

$$A_1 \cap (A_2 \cup A_3).$$

So

$$P(W) = P(A_1)P(A_2 \cup A_3) = p(p + p - p^2) = p(2p - p^2) = p^2(2 - p).$$

17 Bernoulli Trials and Bernoulli Random Variables

A Bernoulli trial has two outcomes: S (success) and F (failure):

$$P(S) = p, \quad P(F) = 1 - p.$$

A random variable $X : \Omega \rightarrow \mathbb{R}$ is called a **Bernoulli random variable** with parameter p if

$$X(\omega) = \begin{cases} 1, & \omega \in S, \\ 0, & \omega \in F. \end{cases}$$

Notation:

$$X \sim \text{Ber}(p).$$

The pmf is

$$p_X(x) = \begin{cases} 1 - p, & x = 0, \\ p, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

17.1 Example

Let $Y(\omega) = 2X(\omega) + 0.5$. Then

$$Y(\omega) = \begin{cases} 0.5, & X = 0, \\ 2.5, & X = 1. \end{cases}$$

Thus

$$P(Y = 0.5) = 1 - p, \quad P(Y = 2.5) = p.$$

17.2 Probability Measure on a Two-Point Space

Let $\Omega = \{0, 1\}$ and define a measure Q by

$$Q(B) = \begin{cases} p, & 1 \in B, 0 \notin B, \\ 1 - p, & 0 \in B, 1 \notin B, \\ 0, & \text{if neither 0 nor 1 is in } B, \\ 1, & \text{if both 0 and 1 are in } B, \end{cases} \quad 0 \leq p \leq 1.$$

This defines a valid probability measure on 2^Ω .

18 Distribution Induced by a Random Variable

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let

$$X : \Omega \rightarrow \mathbb{R}$$

be a random variable. For any Borel set $B \subset \mathbb{R}$,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}.$$

Define the induced probability measure

$$Q(B) = P(X^{-1}(B)).$$

Then Q is the distribution of the random variable X .

Basic Properties of Preimages

For any sets $A, B \subset \mathbb{R}$,

$$X^{-1}(A^c) = (X^{-1}(A))^c, \quad X^{-1}(A \cup B) = X^{-1}(A) \cup X^{-1}(B),$$

$$X^{-1}(A \cap B) = X^{-1}(A) \cap X^{-1}(B).$$

Examples

Example 1: If

$$X(\omega) = \begin{cases} 0, & \omega \notin S, \\ 1, & \omega \in S, \end{cases}$$

then $X^{-1}(\{1\}) = S$ and $X^{-1}(\{0\}) = S^c$.

Example 2: If $B_1 = \{3\}$, then $X^{-1}(B_1) = \{\omega : X(\omega) = 3\}$.

If $B_2 = [-3, 3]$, then $X^{-1}(B_2) = \Omega$ (as shown in notes).

If B_3 is the set of odd integers, then $X^{-1}(B_3) = \emptyset$.

—

19 Repeated Bernoulli Trials

Consider n independent Bernoulli trials, each having two outcomes: Success S with probability p , Failure F with probability $1 - p$.

Total number of possible outcomes:

$$2^n.$$

Examples of outcome strings:

$$SSS \dots S, \quad FSSS \dots S, \quad SFFFF \dots S.$$

Probability of Getting Exactly r Successes

If event $A =$ “exactly r successes”,

$$P(A) = \binom{n}{r} p^r (1 - p)^{n-r}.$$

—

20 Binomial Distribution

Let X be the number of successes in n independent Bernoulli trials with success probability p . Then

$$X \sim \text{Bin}(n, p).$$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Normalization check:

$$\sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} = (p + (1 - p))^n = 1.$$

—

21 Geometric Distribution

Let the Bernoulli trials be repeated until the first success occurs.

Let

$X =$ number of failures before the first success.

Then

$$P(X = k) = (1 - p)^k p, \quad k = 0, 1, 2, \dots$$

This is the geometric distribution with parameter p .

Alternate form (trial index of first success):

$$T = X + 1, \quad P(T = t) = (1 - p)^{t-1} p.$$

—

Indicator Function Example

For $A \subset \Omega$, define

$$I_A(\omega) = \begin{cases} 0, & \omega \notin A, \\ 1, & \omega \in A. \end{cases}$$

Then

$$I_A \sim \text{Ber}(P(A)).$$

—

22 Negative Binomial Distribution

Let

X = number of failures before the r -th success.

Then the pmf is

$$P(X = x) = \binom{x+r-1}{r-1} p^r (1-p)^x, \quad x = 0, 1, 2, \dots$$

Useful identity:

$$\binom{x+r-1}{r-1} = \frac{(x+r-1)(x+r-2) \cdots (x)}{(r-1)!}.$$

Normalization (as in notes):

$$\sum_{x=0}^{\infty} \binom{x+r-1}{r-1} p^r (1-p)^x = p^r \sum_{x=0}^{\infty} \binom{x+r-1}{r-1} (1-p)^x = 1.$$

—

23 Repeated Bernoulli Trials Without Independence

In this case, the success probability changes from trial to trial.

Example: Urn Sampling Without Replacement

Urn contains r red balls and g green balls. Draw one ball without replacement.

$$P(\text{red on 1st}) = \frac{r}{r+g}.$$

For second draw:

$$P(\text{red on 2nd}) = \begin{cases} \frac{r-1}{r+g-1}, & \text{if 1st was red,} \\ \frac{r}{r+g-1}, & \text{if 1st was green.} \end{cases}$$

This leads to the ****Hypergeometric distribution****, because sampling is without replacement.

Random Variable

Ω = Sample space, $\mathcal{F}$ = collection of all events, P .
 $(\Omega, \mathcal{F}, P)$.

A random variable X is defined as a function

$$X : \Omega \rightarrow \mathbb{R}$$

i.e., for every $\omega \in \Omega$, $X(\omega) \in \mathbb{R}$.

Event: $\{\omega : X(\omega) \in [a, b]\}$, $\{\omega : X(\omega) = x\}$, $\{\omega : X(\omega) \in (a, b, c]\}$.

Definition: A random variable is discrete if the set of its values is countable.

Definition: PMF The probability mass function (pmf) of a discrete random variable X is

$$p(x) = P(X = x), \quad p : \mathbb{R} \rightarrow [0, 1].$$

Properties:

$$p(x) \geq 0 \quad \forall x$$

$$\sum_x p(x) = 1$$

$$P\left(\bigcup_x \{X = x\}\right) = 1 \quad (\text{countable additivity})$$

Example 1: $\Omega = \{H, T\}$ $X : \Omega \rightarrow \mathbb{R}$, $X(H) = 1$, $X(T) = 0 \Rightarrow X \sim \text{Ber}(p = P(H))$.

$Y : \Omega \rightarrow \mathbb{R}$, $Y(H) = 0$, $Y(T) = 1 \Rightarrow Y \sim \text{Ber}(q = P(T))$.

$Z : \Omega \rightarrow \mathbb{R}$, $Z(H) = 1$, $Z(T) = 1$.

PMF:

$$p_Z(z) = \begin{cases} P(H) = p, & z = 1 \\ P(T) = q, & z = 0 \\ 0, & \text{otherwise} \end{cases}$$

$T : \Omega \rightarrow \mathbb{R}$, $T(\omega) = 1$ for all $\omega \in \Omega$.

$$p_T(t) = \begin{cases} 1, & t = 1 \\ 0, & \text{otherwise} \end{cases}$$

Example 2: $\Omega = \{1, 2, 3, \dots, 6\}$, $P(\{\omega\}) = 1/6$.

$X(\omega) = \omega$, $Y(\omega) = 6 - \omega$, $Z(\omega) = \omega^2$.

$$p_Z(z) = \begin{cases} 1/6, & z \in \{1, 4, 9, 16, 25, 36\} \\ 0, & \text{otherwise} \end{cases}$$

Example 3: $\Omega = [0, 1]$. $X(\omega) = \omega$. X is not a discrete random variable. Define

$$Y(\omega) = k/n \quad \text{if } k/n \leq \omega < (k+1)/n, \quad k = 0, 1, 2, \dots, n-1.$$

Example: X = Number of customers arriving at a shop between 12:00–12:10 PM.

$$\begin{array}{cccc} x : & 0 & 1 & 2 & 3 \\ p(x) : & 0.3 & 0.2 & 0.4 & 0.1 \end{array}$$

Find $P(0 \leq X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = 0.9$.

Property: For any discrete r.v. X with pmf $p(x)$,

$$P(X \in A) = \sum_{x \in A} p(x).$$

Example: Urn with 2 black, 3 green, 2 red balls. A ball is removed repeatedly until a red ball is drawn. Let X = number of draws necessary.

Similarly, define experiments for green ball and black ball. Let Y and Z be respective number of draws.

Exercise: Find pmfs of X, Y, Z .

Theorem (Poisson limit). Let $\lambda > 0$, $k \geq 1$, $n \geq \lambda/p$, $p = \lambda/n$. Let A_k = “ k successes in n independent Bernoulli(p) trials”.

$$\lim_{n \rightarrow \infty} P(A_k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

Proof steps (matching your notebook):

$$\begin{aligned} \binom{n}{k} p^k (1-p)^{n-k} &= \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{n(n-1) \cdots (n-k+1)}{k!} \frac{\lambda^k}{n^k} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k} \\ &\rightarrow \frac{\lambda^k}{k!} e^{-\lambda}. \end{aligned}$$

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

$$\sum_{k=0}^{\infty} p(k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$$

Thus, $p(k)$ is a pmf.

Definition: The pmf

$$p(x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!}, & x = 0, 1, 2, \dots, \\ 0, & \text{otherwise} \end{cases}$$

is called the Poisson pmf with parameter $\lambda > 0$.

A r.v. Y with this pmf is called Poisson:

$$Y \sim \text{Poi}(\lambda).$$

—

Expected Value of a Discrete Random Variable

Let X have pmf $p(x)$.

$$E(X) = \sum_x x p(x) = \sum_x x P(X = x).$$

Expected Value

Example 1

x	1	2	3	4	5	6
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E(X) = \sum_{x=1}^6 x p(x) = \frac{1 \cdot 1 + 2 \cdot 1 + 3 \cdot 1 + 4 \cdot 1 + 5 \cdot 1 + 6 \cdot 1}{6} = \frac{21}{6} = 3.5$$

X is called a discrete uniform random variable.

More generally, if Y has a discrete uniform distribution on $\{1, 2, \dots, N\}$, then

$$p(i) = \frac{1}{N}, \quad i = 1, 2, \dots, N$$

$$E(Y) = \frac{1}{N}(1 + 2 + \dots + N) = \frac{N+1}{2}$$

Example 2

x	-2	-1	0	1	2	4
$p(x)$	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{2}{7}$	$\frac{1}{7}$

$$E(X) = (-2) \cdot \frac{1}{7} + (-1) \cdot \frac{2}{7} + 0 \cdot \frac{1}{7} + 1 \cdot \frac{2}{7} + 2 \cdot \frac{2}{7} + 4 \cdot \frac{1}{7} = \frac{4}{7}$$

Expected value (“central value”) gives a summary of a random variable or its distribution.

Example 3

(i) If $X \sim \text{Ber}(p)$:

$$E(X) = 0 \cdot (1 - p) + 1 \cdot p = p$$

(ii) If $X \sim \text{Bin}(2, p)$:

$$p(x) = \begin{cases} (1-p)^2 & x = 0 \\ 2p(1-p) & x = 1 \\ p^2 & x = 2 \end{cases}$$

$$E(X) = 0 \cdot (1-p)^2 + 1 \cdot 2p(1-p) + 2p^2 = 2p$$

(iii) If $X \sim \text{Bin}(n, p)$:

$$\begin{aligned} E(X) &= \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} \\ &= np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} (1-p)^{(n-1)-(x-1)} \end{aligned}$$

Let $y = x - 1$:

$$= np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y (1-p)^{(n-1)-y} = np(1) = np$$

(iv) If $X \sim \text{Poi}(\lambda)$:

$$E(X) = \sum_{x=0}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!} = \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda$$

Exercise: Let $X \sim \text{NB}(r, p)$. Find $E(X)$.

Further Examples

Example

x	-1	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Let $Y = X^2$. Then:

$$E(Y) = \sum_y y p(Y = y)$$

$$p_Y(y) = \begin{cases} \frac{1}{4}, & y = 0 \\ \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, & y = 1 \\ \frac{1}{4}, & y = 4 \\ 0, & \text{otherwise} \end{cases}$$

$$E(Y) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = \frac{3}{4}$$

$$E(X^2) = \sum_x x^2 p(x)$$

Definition

$$E[h(X)] = \sum_x h(x) p(x)$$

Proposition

Let $a, b \in \mathbb{R}$. Then

$$E(aX + b) = aE(X) + b$$

Proof:

$$E(aX + b) = \sum_x (ax + b)p(x) = a \sum_x xp(x) + b \sum_x p(x) = aE(X) + b$$

Exercise

Let

$$f(x) = \sum_{n=0}^N a_n x^n$$

Show that

$$E[f(X)] = \sum_{n=0}^N a_n E(X^n)$$

Cautionary Examples

Example

Suppose Z is a r.v. with pmf

$$p(Z) = \frac{1}{2^n}, \quad Z = 2^n, \quad n = 1, 2, \dots$$

$$E(Z) = \sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} 1 = \infty$$

Expectation does **not** exist.

Example

Let

$$b(y) = \frac{c}{y^2}, \quad y = 1, 2, \dots$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

So choose

$$c = \frac{6}{\pi^2}$$

Then $b(y)$ is a pmf.

Continuation

Let

$$T = (-1)^n 2^n, \quad n = 1, 2, \dots$$

with probability $1/2^n$.

$$p_T(t) = \frac{1}{2^n}, \quad t = (-1)^n 2^n$$

$$E(T) = \sum_{n=1}^{\infty} (-1)^n 2^n \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} (-1)^n$$

So $E(T)$ does not exist.

Remarks

- 1) If a random variable takes only finitely many values, then its expectation always exists.
- 2) If X takes infinitely many values but $0 \leq x \leq k$, then

$$E(X) = \sum_x x p(x) \leq \sum_x k p(x) = k < \infty$$

Heavy Tails and Non-Existence of Expectation

If X takes arbitrarily large values with positive probability, then

$$\sum_x x p(x)$$

may diverge.

Tail probabilities:

$$P(X > x) = \sum_{s > x} p(s)$$

If the distribution has “heavy tails”, then the expectation may not exist (e.g. Cauchy distribution).

Example

A store has purchased 3 items at Rs. 500 per item. Each item will be sold for Rs. 1000 per piece. Unsold items at the end of the period will be bought back by the company for Rs. 200 per piece.

Let X = number of pieces sold.

$$P(0) = 0.1, \quad P(1) = 0.2, \quad P(2) = 0.3, \quad P(3) = 0.4.$$

- (a) Find the expected profit for the store.

Let Y denote the actual profit. Selling a piece gives profit Rs. $(1000 - 500) = 500$; an unsold piece gives loss Rs. $(200 - 500) = -300$.

$$Y = 500X - 300(3 - X) = 800X - 900.$$

$$E(Y) = 800E(X) - 900.$$

Now

$$E(X) = \sum_{x=0}^3 x p(x) = 0 \cdot 0.1 + 1 \cdot 0.2 + 2 \cdot 0.3 + 3 \cdot 0.4 = 2.$$

Hence

$$E(Y) = 800 \cdot 2 - 900 = 700.$$

Example

Two different random variables can have the same expected value.

(i) Let X have

$$\begin{array}{c|cccc} x & 1 & 2 & 3 & 4 \\ \hline p(x) & 0.2 & 0.6 & 0.2 & 0 \end{array} \quad \Rightarrow \quad E(X) = 2.$$

(ii) Let Y have

$$\begin{array}{c|cccc} y & 1 & 2 & 3 & 4 \\ \hline p_Y(y) & 0.4 & 0.3 & 0.2 & 0.1 \end{array} \quad \Rightarrow \quad E(Y) = 2.$$

Thus X and Y have the same expectation but different distributions.

Variance and Standard Deviation

Definition

Let X be a discrete r.v. with mean $\mu = E(X)$ and pmf $p(x)$. The variance of X is

$$V(X) = E[(X - \mu)^2].$$

The standard deviation of X is

$$\sqrt{\text{Var}(X)} = \sqrt{V(X)} \quad (\text{always } \geq 0).$$

For a discrete r.v.,

$$V(X) = \sum_x (x - E(X))^2 p(x).$$

Example

Using the two distributions above:

$$\begin{aligned} V(X) &= (1 - 2)^2 p(1) + (2 - 2)^2 p(2) + (3 - 2)^2 p(3) = 0.4, \\ V(Y) &= (1 - 2)^2 p_Y(1) + (2 - 2)^2 p_Y(2) + (3 - 2)^2 p_Y(3) + (4 - 2)^2 p_Y(4) = 1. \end{aligned}$$

Remarks

1)

$$E(X - \mu) = E(X) - \mu = 0.$$

2) $E|X - \mu|$ is the mean of the absolute deviations from the mean.

3) Shortcut formula:

$$V(X) = E[(X - \mu)^2] = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - \mu^2 = E(X^2) - [E(X)]^2.$$

Variance of a Linear Transformation

$$V(aX + b) = E[(aX + b)^2] - [E(aX + b)]^2.$$

Now

$$E[(aX + b)^2] = E(a^2X^2 + 2abX + b^2) = a^2E(X^2) + 2abE(X) + b^2,$$

$$[E(aX + b)]^2 = [aE(X) + b]^2 = a^2[E(X)]^2 + 2abE(X) + b^2.$$

Hence

$$V(aX + b) = a^2E(X^2) - a^2[E(X)]^2 = a^2V(X).$$

Example: Bernoulli

If $X \sim \text{Ber}(p)$, then

$$E(X) = p,$$

$$E(X^2) = 0^2p(0) + 1^2p(1) = p,$$

$$V(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1 - p).$$

Example: Binomial

If $X \sim \text{Bin}(n, p)$ with

$$p(x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n,$$

and 0 otherwise.

Consider the factorial moment

$$E[X(X - 1)] = \sum_{x=0}^n x(x - 1) \binom{n}{x} p^x (1 - p)^{n-x}.$$

Using

$$x(x - 1) \binom{n}{x} = n(n - 1) \binom{n - 1}{x - 1} \frac{x - 1}{n - 1} = n(n - 1) \binom{n - 2}{x - 2},$$

we get

$$E[X(X-1)] = n(n-1)p^2 \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} (1-p)^{(n-2)-(x-2)} = n(n-1)p^2.$$

Also,

$$E(X^2 - X) = E[X(X-1)],$$

so

$$E(X^2) - E(X) = E[X(X-1)].$$

Hence

$$E(X^2) = E[X(X-1)] + E(X) = n(n-1)p^2 + np.$$

Therefore

$$V(X) = E(X^2) - [E(X)]^2 = n(n-1)p^2 + np - (np)^2 = np(1-p).$$

Poisson and Negative Binomial

If $X \sim \text{Poi}(\lambda)$, then (parameter λ)

$$E(X) = \lambda, \quad V(X) = \lambda.$$

If $X \sim \text{NB}(r, p)$ (number of failures before r successes, with probability of success p), then

$$E(X) = \frac{r(1-p)}{p}, \quad V(X) = \frac{r(1-p)}{p^2}.$$

Jointly Distributed Random Variables

Remarks

For a joint pmf $p(x, y)$,

$$p(x, y) \geq 0 \quad \forall x, y,$$

$$\sum_x \sum_y p(x, y) = 1.$$

If $A \subset \mathbb{R}^2$, then

$$P((X, Y) \in A) = \sum_{(x, y) \in A} p(x, y).$$

Example

A petrol pump station has two pumps for petrol and one for diesel.

Let X = number of pumps for petrol that are in use, Y = number of pumps for diesel that are in use.

The joint pmf is:

	$y = 0$	$y = 1$	$p_X(x)$
$x = 0$	0.20	0.05	0.25
$x = 1$	0.10	0.15	0.25
$x = 2$	0.20	0.30	0.50
$p_Y(y)$	0.50	0.50	

$$P(X \leq 1) = P(X \leq 1, Y \in \mathbb{R}) = \sum_{x \leq 1} \sum_{y=0}^1 p(x, y) = 0.50.$$

$$P(X + Y = 1) = \sum_{x+y=1} p(x, y) = 0.15.$$

$$P(X = 0) = p(0, 0) + p(0, 1) = 0.25, \quad P(X = 1) = 0.25, \quad P(X = 2) = 0.50.$$

Marginal pmfs

The marginal pmf of X , denoted $p_X(x)$, is

$$p_X(x) = \sum_y p(x, y).$$

The marginal pmf of Y , denoted $p_Y(y)$, is

$$p_Y(y) = \sum_x p(x, y).$$

Conditional Distributions

Conditional pmf of X given $Y = y$

$$p_{X|Y}(x|y) = P(X = x | Y = y) = \frac{p(x, y)}{p_Y(y)}, \quad p_Y(y) > 0.$$

(read as “ p of x given y ”)

Conditional probability of $X = x$ given $Y = y$

$$p_{X|Y}(x|y) = \frac{p(x, y)}{p_Y(y)}, \quad p_Y(y) > 0.$$

Similarly, conditional probability of $Y = y$ given $X = x$:

$$p_{Y|X}(y|x) = \frac{p(x, y)}{p_X(x)}, \quad p_X(x) > 0.$$

Conditional pmf of X given $Y = 0$

$$p_{X|Y}(x|0) = \frac{p(x, 0)}{p_Y(0)} = \frac{p(x, 0)}{0.5}.$$

$$p_{X|Y}(x|0) = \begin{cases} 0.4, & x = 0, \\ 0.2, & x = 1, \\ 0.4, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

$$p_{Y|X}(y|x), \quad y = 0, 1, 2.$$

Definition

Two discrete random variables with joint pmf $p(x, y)$ and marginal pmfs $p_X(x)$ and $p_Y(y)$ are *independent* if

$$p(x, y) = p_X(x)p_Y(y) \quad \forall x, y.$$

If this does not hold, X and Y are *dependent*.

Example: Two Tosses of a Fair Die

Let X be the outcome of the first toss and Y the outcome of the second toss.

$$p_X(x) = \frac{1}{6}, \quad x = 1, 2, 3, 4, 5, 6,$$
$$p_Y(y) = p_X(y).$$

Since the two tosses are independent:

$$p(x, y) = p_X(x)p_Y(y) = \frac{1}{36}, \quad x, y = 1, 2, \dots, 6.$$

Exercise

Let $Z = X + Y$. Find the pmf of Z .

Possible values:

$$Z = 2, 3, 4, \dots, 12.$$

$$P(Z = z) = P(X + Y = z) = \sum_{x+y=z} p(x, y).$$

For example:

$$P(Z = 2) = p(1, 1) = \frac{1}{36},$$

$$P(Z = 3) = p(1, 2) + p(2, 1) = \frac{2}{36}.$$

Expectation

$$E(X) = \sum_x x p(x) = \sum_x \sum_y x p(x, y).$$

Definition

Let $h(x, y)$ be a real-valued function on $\mathbb{R}^2$. Then

$$E[h(X, Y)] = \sum_x \sum_y h(x, y) p(x, y).$$

Remark

$$E[h(X, Y)] \text{ exists iff } E(|h(X, Y)|) < \infty.$$

Example: Drawing Chits Without Replacement

Consider 5 chits numbered 1, 2, 3, 4, 5. Two chits are drawn randomly (without replacement) one after another.

Let X and Y denote the numbers on the first and second chit respectively.

$$\begin{aligned} p(x, y) &= \frac{1}{20}, & x, y \in \{1, 2, 3, 4, 5\}, \ x \neq y, \\ p(x, y) &= 0, & \text{otherwise.} \end{aligned}$$

Absolute Value Expectation

$$E(|X - Y|) = \sum_{x, y} |x - y| p(x, y) = \frac{15}{20}.$$

Variance via Joint Distribution

$$V(X) = E[(X - \mu_X)^2] = E(X^2) - [E(X)]^2.$$

$$E(X^2) = \sum_{x, y} x^2 p(x, y), \quad E(X) = \sum_{x, y} x p(x, y).$$

Covariance

Definition

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)].$$

Remarks

1) $\text{Cov}(X, X) = V(X)$.

2)

$$\text{Cov}(X, Y) = E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y].$$

Thus,

$$\text{Cov}(X, Y) = E(XY) - \mu_X\mu_Y.$$

Interpretation of Covariance

If large values of X tend to appear with large values of Y (positive relationship), then

$$\text{Cov}(X, Y) > 0.$$

If $\text{Cov}(X, Y) < 0$, large values of X tend to occur with small values of Y .
(Diagrams omitted but represent upward/downward linear trend.)

Revision

$$E(X) = \sum_x x p(x)$$

$$V(X) = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$$

$E(X)$ exists iff $E(|X|) < \infty$; hence $E(|X|) = \sum |x|p(x) < \infty$.

Moments

Moments describe the shape of probability distributions.

Types of Moments

(i) **Raw Moments (Moments about origin)** The k th raw moment is

$$\mu'_k = E(X^k), \quad \text{if it exists.}$$

The 1st raw moment:

$$\mu'_1 = E(X),$$

which is the mean of the distribution.

(ii) **Central Moments (Moments about mean)** The k th central moment is

$$\mu_k = E[(X - \mu)^k].$$

The 2nd central moment gives measure of spread (variance).

Note: k th raw moment exists iff k th central moment exists.

Moment Generating Function (mgf)

The mgf of a random variable X is defined as

$$M_X(t) = E(e^{tX}) = \sum_x e^{tx} p(x).$$

Remark:

$$M_X(0) = 1.$$

Examples of mgf

Example 1: Poisson Distribution

If $X \sim \text{Poi}(\lambda)$,

$$M_X(t) = E(e^{tX}) = \sum_{x=0}^{\infty} e^{tx} e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} = e^{\lambda(e^t-1)}.$$

Example 2: Binomial Distribution

If $X \sim \text{Bin}(n, p)$,

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x} = (1-p + pe^t)^n. \end{aligned}$$

Example 3: Geometric Distribution

Case I: Number of trials until 1st success Let $X \sim \text{Geom}(p)$,

$$M_X(t) = \sum_{x=1}^{\infty} e^{tx} p(1-p)^{x-1} = \frac{pe^t}{1 - (1-p)e^t}.$$

Case II: Number of failures before 1st success

$$M_X(t) = \sum_{x=0}^{\infty} e^{tx} p(1-p)^x = \frac{p}{1 - e^t(1-p)}.$$

Example 4: Bernoulli Distribution

If $X \sim \text{Ber}(p)$,

$$M_X(t) = \sum_{x=0}^1 e^{tx} p(x) = p(1)e^t + p(0) = pe^t + (1-p).$$

Example 5: Negative Binomial Distribution

If $X \sim \text{NB}(r, p)$, number of failures before r th success:

$$M_X(t) = \sum_{k=0}^{\infty} e^{tk} \binom{k+r-1}{k} p^r (1-p)^k.$$

Using series expansion,

$$M_X(t) = p^r \left[\sum_{k=0}^{\infty} \binom{k+r-1}{k} (e^t(1-p))^k \right],$$

$$M_X(t) = \left(\frac{p}{1 - e^t(1-p)} \right)^r.$$

Theorem

Suppose X is a discrete random variable with pmf $p(x)$. If $M_X(t)$ exists for all $t \in [-\delta, \delta]$, then the k th moment exists and is given by:

$$E(X^k) = M_X^{(k)}(0).$$

Proof Sketch

$$e^{tx} = \sum_{n=0}^{\infty} \frac{(tx)^n}{n!},$$

$$\begin{aligned} M_X(t) &= E(e^{tX}) = E \left[1 + tX + \frac{t^2 X^2}{2!} + \cdots \right] \\ &= M_X(0) + tE(X) + \frac{t^2}{2!} E(X^2) + \cdots \end{aligned}$$

Differentiating:

$$M_X'(t) = E(X) + tE(X^2) + \frac{t^2}{2!} E(X^3) + \cdots$$

Similarly,

$$M_X^{(k)}(t) = \sum_{n=k}^{\infty} \frac{t^{n-k}}{(n-k)!} E(X^n)$$

Thus,

$$M_X^{(k)}(0) = E(X^k).$$

Example: Using mgf of Binomial

We have

$$M_X(t) = (1 - p + pe^t)^n.$$

1st derivative:

$$\begin{aligned}M'_X(t) &= n(1 - p + pe^t)^{n-1} e^t p \\M'_X(0) &= np.\end{aligned}$$

2nd derivative:

$$\begin{aligned}M''_X(t) &= n(n-1)(1 - p + pe^t)^{n-2} (e^t p)^2 + n(1 - p + pe^t)^{n-1} e^t p \\M''_X(0) &= n(n-1)p^2 + np.\end{aligned}$$

Thus,

$$E(X^2) = n(n-1)p^2 + np,$$

and variance:

$$V(X) = E(X^2) - [E(X)]^2 = np(1 - p).$$

Properties of mgf

(1) For $a \neq 0$, $a \in \mathbb{R}$ such that $at \in (-\delta, \delta)$,

$$M_{aX}(t) = M_X(at).$$

(2) If X and Y are independent random variables, then

$$M_{X+Y}(t) = M_X(t) M_Y(t).$$

(3) Let X and Y be discrete random variables with joint pmf $p(x, y)$. Suppose $E(X)$ and $E(Y)$ exist. Then

$$E(XY) = E(X)E(Y).$$

Proof:

$$E(XY) = \sum_x \sum_y xy p(x, y) = \sum_x xp_X(x) \sum_y yp_Y(y) = E(X)E(Y).$$

(4) Let X_1 and X_2 be independent r.v.'s such that $X_1 \sim \text{Poi}(\lambda_1)$ and $X_2 \sim \text{Poi}(\lambda_2)$. Then

$$M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t) = e^{\lambda_1(e^t-1)}e^{\lambda_2(e^t-1)} = e^{(\lambda_1+\lambda_2)(e^t-1)}.$$

(5) If X and Y are independent:

$$\begin{aligned}M_{X+Y}(t) &= M_X(t)M_Y(t), \\E[f(X)g(Y)] &= E[f(X)]E[g(Y)], \\E(XY) &= E(X)E(Y).\end{aligned}$$

(6) If X_1, X_2 are independent Poisson with λ_1, λ_2 , then $X_1 + X_2$ is Poisson with parameter $\lambda_1 + \lambda_2$.

Properties of mgf (continued)

Fact 1: mgf does not determine the distribution of the r.v. in all cases. **Fact 2:** In this course we will consider mgf determines the distribution.

Theorem

Let $X_1, X_2, \dots, X_n$ be independent with:

(a) $X_i \sim \text{Poi}(\lambda_i)$, $i = 1, 2, \dots, n$. Then

$$X_1 + \dots + X_n \sim \text{Poi}(\lambda_1 + \dots + \lambda_n).$$

(b) $X_i \sim \text{Bin}(n_i, p)$. Then

$$\sum_{i=1}^n X_i \sim \text{Bin}\left(\sum n_i, p\right).$$

(c) $X_i \sim \text{Geom}(p)$,

$$\sum_{i=1}^n X_i \sim \text{NB}(n, p).$$

Covariance

Given joint pmf $p(x, y)$,

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X\mu_Y.$$

If X and Y are independent,

$$\text{Cov}(X, Y) = 0.$$

But the converse is not true.

Example

$$p(x, y) = \frac{1}{4}, \quad (x, y) \in \{(-4, 4), (4, -1), (2, 2), (-2, 2)\},$$

$$p(x, y) = 0 \text{ otherwise.}$$

Check $\text{Cov}(X, Y) = 0$.

If $(x - \mu_X)$ and $(y - \mu_Y)$ tend to be both positive or both negative, then $\text{Cov}(X, Y) > 0$.
(Diagrams represent positive negative relationships.)

Theorem

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y).$$

Proof:

$$\mu_{aX+b} = aE(X) + b, \quad \mu_{cY+d} = cE(Y) + d,$$

$$\text{Cov}(aX + b, cY + d) = E[(aX + b - \mu_{aX+b})(cY + d - \mu_{cY+d})] = ac \text{Cov}(X, Y).$$

Remark: $\text{Cov}(X, X) = \text{Var}(X)$.

Recall:

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Correlation

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

where σ_X, σ_Y are standard deviations.

Theorem

If a, c are both positive or both negative, then

$$\rho(aX + b, cY + d) = \rho(X, Y).$$

Proof:

$$\rho(aX + b, cY + d) = \frac{\text{Cov}(aX + b, cY + d)}{\sigma_{aX+b} \sigma_{cY+d}} = \frac{ac \text{Cov}(X, Y)}{|a| \sigma_X |c| \sigma_Y} = \rho(X, Y).$$

Theorem

(a) $\rho(X, Y) \in [-1, 1]$.

(b) If $Y = aX + b$ for constants $a \neq 0$, then $\rho(X, Y) = \pm 1$.

(c) If $\rho(X, Y) = \pm 1$, then $Y = aX + b$ for some $a \neq 0$.

(Diagrams indicate negative correlation, perfect positive, positive, no correlation.)

Theorem

$$V(X + Y) = V(X) + V(Y) + 2\text{Cov}(X, Y).$$

Proof:

$$\begin{aligned} V(X + Y) &= E[(X + Y - E(X + Y))^2] \\ &= E[(X - E(X))^2] + E[(Y - E(Y))^2] + 2E[(X - E(X))(Y - E(Y))] \\ &= V(X) + V(Y) + 2\text{Cov}(X, Y). \end{aligned}$$

Corollary: If X and Y independent,

$$V(X + Y) = V(X) + V(Y).$$

Theorem: Variance of Sum

(a)

$$V\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n V(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j).$$

(b) If $X_1, \dots, X_n$ independent,

$$V\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n V(X_i).$$

Theorem: Bounds on Correlation

(a) $\rho(X, Y) \in [-1, 1]$.

Proof:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)].$$

Consider:

$$\begin{aligned} 0 &\leq E\left[\left(\frac{X - \mu_X}{\sigma_X} + \frac{Y - \mu_Y}{\sigma_Y}\right)^2\right] \\ &= E\left(\frac{(X - \mu_X)^2}{\sigma_X^2} + \frac{(Y - \mu_Y)^2}{\sigma_Y^2} + 2\frac{(X - \mu_X)(Y - \mu_Y)}{\sigma_X \sigma_Y}\right) \\ &= 1 + 1 + 2\rho(X, Y). \end{aligned}$$

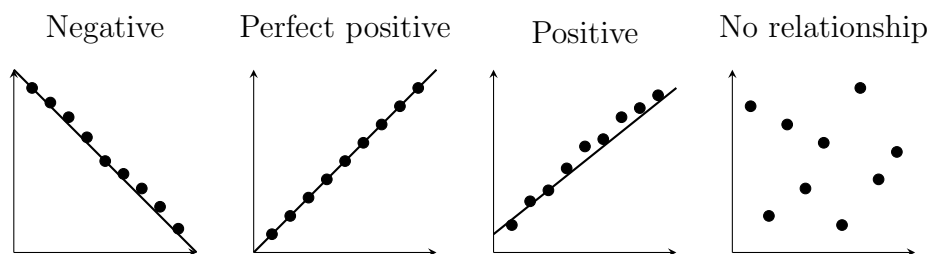
Thus,

$$\rho(X, Y) \geq -1.$$

Similarly, considering

$$E\left[\left(\frac{X - \mu_X}{\sigma_X} - \frac{Y - \mu_Y}{\sigma_Y}\right)^2\right]$$

gives $\rho(X, Y) \leq 1$.



24 Continuous Random Variable

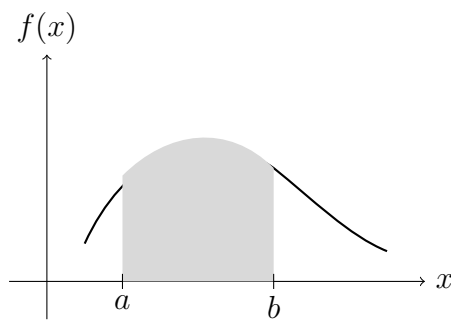
Defn. Probability density function (pdf): A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

1. $f(x) \geq 0, \forall x \in \mathbb{R}$,
2. $\int_{-\infty}^{\infty} f(x) dx = 1$

is called a pdf.

Defn. A continuous random variable X has pdf $f(x)$ if for any $a \leq b$

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$



Remarks.

1. A continuous random variable takes values in an interval or union of intervals.
2. $P(X = a) = \int_a^a f(x) dx = 0$.
3. For any $a < b$,

$$P(a < X \leq b) = P(a \leq X \leq b) = P(a \leq X < b) = P(a < X < b).$$

Examples.

1. Pick a number at random from $[0, 1]$.
2. Height/weight of a randomly chosen person.
3. Amount of rainfall in a month.

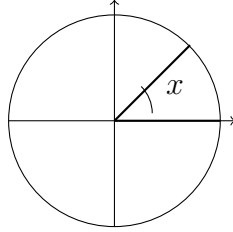
Example. There is an imperfection on a pipe. The direction of imperfection is random. Let X = angle measured clockwise to the location of the imperfection from some reference line joining the centre of the wall.

Then X takes values in $[0, 360]$.

$$f(x) = \begin{cases} \frac{1}{360}, & 0 \leq x \leq 360, \\ 0, & \text{otherwise.} \end{cases}$$

Check:

$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 0 dx + \int_0^{360} \frac{1}{360} dx + \int_{360}^{\infty} 0 dx = 0 + \frac{1}{360} [x]_0^{360} + 0 = 1.$$

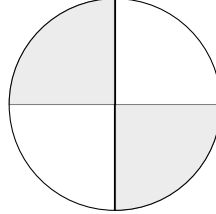


(a) Probability that the imperfection is between 90° and 180° :

$$P(90^\circ \leq X \leq 180^\circ) = \int_{90}^{180} f(x) dx = \int_{90}^{180} \frac{1}{360} dx = \frac{1}{360} (180 - 90) = \frac{1}{4}.$$

(b) Probability that the imperfection is within 90° of the reference line:

$$P(X \in [0, 90] \cup [270, 360]) = P(X \in [0, 90]) + P(X \in [270, 360]) = \int_0^{90} \frac{1}{360} dx + \int_{270}^{360} \frac{1}{360} dx = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

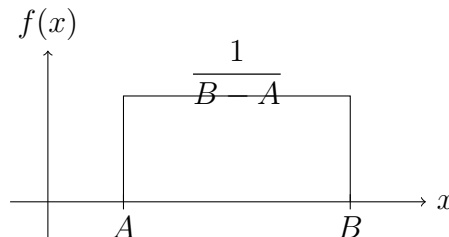


Uniform Distribution

Defn. A continuous random variable X is said to have a uniform distribution on $[A, B]$ if its pdf is

$$f(x) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B, \\ 0, & \text{otherwise.} \end{cases}$$

Notation: $X \sim U([A, B])$.



Example. Pick a number at random from $[0, 1]$.

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

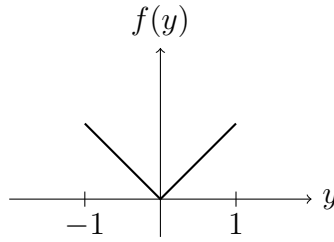
$$P(X \geq \frac{2}{3}) = \int_{2/3}^{\infty} f(x) dx = \int_{2/3}^1 1 dx = 1 - \frac{2}{3} = \frac{1}{3}.$$

Example. Y is a random variable with pdf

$$f(y) = \begin{cases} |y|, & -1 \leq y \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

i.e.

$$f(y) = \begin{cases} -y, & -1 \leq y \leq 0, \\ y, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$



$$P\left(-\frac{1}{3} \leq Y \leq \frac{2}{3}\right) = \int_{-1/3}^0 (-y) dy + \int_0^{2/3} y dy = \left[-\frac{y^2}{2}\right]_{-1/3}^0 + \left[\frac{y^2}{2}\right]_0^{2/3} = \frac{1}{18} + \frac{4}{18} = \frac{5}{18}.$$

We find $P(Y \leq y) = \int_{-\infty}^y f(x) dx$.

(i) For $y \leq -1$,

$$P(Y \leq y) = \int_{-\infty}^y f(x) dx = 0.$$

(ii) For $-1 \leq y \leq 0$,

$$P(Y \leq y) = \int_{-1}^y (-x) dx = \left[-\frac{x^2}{2}\right]_{-1}^y = \frac{1}{2} - \frac{y^2}{2}.$$

(iii) For $0 \leq y \leq 1$,

$$P(Y \leq y) = \int_{-1}^0 (-x) dx + \int_0^y x dx = \frac{1}{2} + \frac{y^2}{2}.$$

(iv) For $y \geq 1$,

$$P(Y \leq y) = 1.$$

Define $F(y) = P(Y \leq y)$. Then $F(y)$ is an increasing function of y and

$$F(y) = \begin{cases} 0, & y \leq -1, \\ \frac{1}{2} - \frac{y^2}{2}, & -1 \leq y \leq 0, \\ \frac{1}{2} + \frac{y^2}{2}, & 0 \leq y \leq 1, \\ 1, & 1 \leq y. \end{cases}$$

Cumulative Distribution Function (cdf)

Defn. Let X be any random variable defined on $(\Omega, \mathcal{F}, P)$. The cumulative distribution function (cdf) of X is

$$F(x) = P(X \leq x), \quad x \in \mathbb{R}.$$

Note: for $a < b$,

$$P(a < X \leq b) = F(b) - F(a).$$

Properties of cdf / Remarks.

1. F is increasing in x , i.e. $F(x_1) \leq F(x_2)$ whenever $x_1 \leq x_2$.
2. $\lim_{x \rightarrow -\infty} F(x) = 0$.
3. $\lim_{x \rightarrow \infty} F(x) = 1$.
4. F is right-continuous:

$$\lim_{x \downarrow x_0} F(x) = F(x_0), \quad \forall x_0 \in \mathbb{R}.$$

Remark. If X is continuous, then F is continuous.

Proof of (2): $\lim_{x \rightarrow -\infty} F(x) = 0$.

Let $x_n \downarrow -\infty$, e.g. $x_n = -n$. Then

$$\{X \leq x_{n+1}\} \subset \{X \leq x_n\} \quad \text{for all } n,$$

so the sequence of events $B_n = \{X \leq x_n\}$ is decreasing. Hence

$$\lim_{n \rightarrow \infty} F(x_n) = \lim_{n \rightarrow \infty} P(B_n) = P\left(\bigcap_{n=1}^{\infty} B_n\right).$$

But $\bigcap_{n=1}^{\infty} B_n = \emptyset$, so

$$\lim_{n \rightarrow \infty} F(x_n) = P(\emptyset) = 0.$$

Proofs of properties (continued)

Proof of (3). $\lim_{x \rightarrow \infty} F(x) = 1$. Proof (sketch from notes): write $a_n \uparrow \infty$ (for example $a_n = n$). Then the events $A_n = \{X \leq a_n\}$ satisfy $A_n \uparrow \Omega$, so

$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = P(\Omega) = 1,$$

hence $\lim_{x \rightarrow \infty} F(x) = 1$.

Proof of (4). F is right-continuous. Fix $x \in \mathbb{R}$. Let $x_n \downarrow x$ (decreasing to x). Put $C_n = \{X \leq x_n\}$. Then $C_n \downarrow \{X \leq x\}$, so $P(C_n) \rightarrow P(X \leq x) = F(x)$. Thus $\lim_{n \rightarrow \infty} F(x_n) = F(x)$ and F is right-continuous.

Remark. If X is a continuous random variable (i.e. has a pdf), then F is continuous (no jumps).

Relationship between pdf and cdf

Let X be a continuous random variable with pdf $f(x)$ and cdf $F(x)$. Then

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy,$$

and, when F is differentiable at x , $F'(x) = f(x)$.

Proposition. If F is the cdf of a continuous random variable then F is a continuous function (and $F(x) = \int_{-\infty}^x f(y) dy$ for all x).

Example. Let $X \sim U(A, B)$ with pdf

$$f(x) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B, \\ 0, & \text{otherwise.} \end{cases}$$

Then the cdf is

$$F(x) = \begin{cases} 0, & x < A, \\ \frac{x-A}{B-A}, & A \leq x \leq B, \\ 1, & x > B. \end{cases}$$

(The cdf is continuous and piecewise linear.)

Note. The cdf is defined for every random variable (continuous or discrete).

Examples of cdf and pdf relationships

If $f(x)$ is a pdf of X and we modify it at a single point x_0 (for example set $g(x) = f(x)$ for $x \neq x_0$ and $g(x_0) = f(x_0) + 5$), then for any $a < b$,

$$\int_a^b g(x) dx = \int_a^b f(x) dx,$$

so the integrals (probabilities of intervals) are unchanged by changing values on a set of measure zero.

Example (from notes). The cdf

$$F(x) = \begin{cases} 0, & x < -1, \\ \frac{1}{2} - \frac{x^2}{2}, & -1 \leq x < 0, \\ \frac{1}{2} + \frac{x^2}{2}, & 0 \leq x < 1, \\ 1, & x \geq 1, \end{cases}$$

has pdf (where it exists)

$$f(x) = \begin{cases} -x, & -1 < x < 0, \\ x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Example (discrete). Let $X \sim \text{Bern}(p)$ (Bernoulli). Then the cdf $F(x) = P(X \leq x)$ satisfies

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & x \geq 1, \end{cases}$$

an increasing step function. (This shows the cdf of a discrete variable is an increasing step function.)

Exponential distribution

Definition. Let $\lambda > 0$. A random variable X has an *exponential distribution* with parameter λ if its pdf is

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the cdf:

$$F(x) = \int_{-\infty}^x f(t) dt = \begin{cases} 0, & x \leq 0, \\ 1 - e^{-\lambda x}, & x > 0. \end{cases}$$

Example (application). Number of radioactive particles remaining at time t : $N_t = N_0 e^{-\lambda t}$ for some $\lambda > 0$ (decay model).

Example (Discrete Random Variable)

Given

$$x : -0.5, 0.25, 3.3, \quad p(x) : 0.1, 0.7, 0.2$$

Find the cdf $F(x)$.

$$F(-0.5) = P(X \leq -0.5) = \sum_{x \leq -0.5} p(x) = p(-0.5) = 0.1$$

$$F(0.25) = \sum_{x \leq 0.25} p(x) = p(-0.5) + p(0.25) = 0.8$$

$$F(3.3) = 1$$



Note. In discrete case,

$$p(x) = F(x) - F(x^-), \quad \forall x,$$

where $F(x^-) = \lim_{y \uparrow x} F(y)$.

Continuous Case

Let X be a continuous random variable with cdf $F(x)$ and pdf $f(x)$:

$$F(x) = \int_{-\infty}^x f(y) dy,$$

$$f(x) = F'(x) \quad \text{where } F'(x) \text{ exists.}$$

Example: Transformation

Let $Y = X^2$.

For $y > 0$:

$$G(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$$

$$G(y) = F(\sqrt{y}) - F(-\sqrt{y})$$

Differentiating:

$$G'(y) = f(\sqrt{y}) \frac{1}{2\sqrt{y}} + f(-\sqrt{y}) \frac{1}{2\sqrt{y}}$$

Thus the pdf of $Y = X^2$ is

$$f_Y(y) = \frac{f(\sqrt{y}) + f(-\sqrt{y})}{2\sqrt{y}}, \quad y > 0,$$

$$f_Y(y) = 0, \quad y \leq 0.$$

Complete Proof (from notes)

Let $g(y) = f_Y(y)$.

For $y > 0$, consider

$$\int_0^y g(z) dz = \int_0^y \frac{f(\sqrt{z}) + f(-\sqrt{z})}{2\sqrt{z}} dz.$$

Let $\sqrt{z} = u \Rightarrow dz = 2u du$:

$$\int_0^{\sqrt{y}} f(u) du + \int_0^{\sqrt{y}} f(-u) du = F(\sqrt{y}) - F(-\sqrt{y}) = G(y).$$

Example

Let $X \sim U(0, 1)$.

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$Y = -\frac{1}{\lambda} \ln(1 - X), \quad \lambda > 0.$$

$$\begin{aligned} G(y) &= P(Y \leq y) = P\left(-\frac{1}{\lambda} \ln(1 - X) \leq y\right) \\ &= P(\ln(1 - X) \geq -\lambda y) = P(1 - X \geq e^{-\lambda y}) \\ &= P(X \leq 1 - e^{-\lambda y}) = F(1 - e^{-\lambda y}) = 1 - e^{-\lambda y}. \end{aligned}$$

Thus $Y \sim \text{Exp}(\lambda)$.

General Transformation Theorem

Let ϕ be a differentiable strictly monotone function on interval I . Let X be a continuous r.v. with pdf $f(x)$ and $Y = \phi(X)$. Then

$$g(y) = f(\phi^{-1}(y)) \left| \frac{d}{dy} \phi^{-1}(y) \right|, \quad y \in \phi(I).$$

Remarks.

$$g(y) dy = f(x) dx.$$

Proof (from notes)

Let F and G be the cdf's of X and Y .

If ϕ is strictly increasing:

$$G(y) = P(Y \leq y) = P(\phi(X) \leq y) = P(X \leq \phi^{-1}(y)) = F(\phi^{-1}(y)).$$

$$G'(y) = f(\phi^{-1}(y)) \frac{d}{dy} \phi^{-1}(y).$$

If ϕ is strictly decreasing:

$$G(y) = P(Y \leq y) = P(\phi(X) \leq y) = P(X \geq \phi^{-1}(y)) = 1 - F(\phi^{-1}(y)),$$

$$G'(y) = f(\phi^{-1}(y)) \left| \frac{d}{dy} \phi^{-1}(y) \right|.$$

Example

Let $X \sim \exp(\lambda)$, $\lambda > 0$. Let $\beta > 0$ and $Y = X^{1/\beta}$.

$$y = x^{1/\beta}, \quad x = y^\beta$$
$$\frac{dx}{dy} = \beta y^{\beta-1}$$

Since

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0,$$

the pdf of Y is:

$$g(y) = f(y^\beta) |\beta y^{\beta-1}| = \lambda e^{-\lambda y^\beta} \beta y^{\beta-1}, \quad y > 0,$$
$$g(y) = 0, \quad y \leq 0.$$

25 Linear Transformations of a Random Variable

25.1 Example

Let X have pdf f , and let $a, b \in \mathbb{R}$ with $a \neq 0$. Define

$$Y = aX + b.$$

Then the pdf of Y is

$$g(y) = \frac{1}{|a|} f\left(\frac{y-b}{a}\right), \quad -\infty < y < \infty.$$

26 Bernoulli Example from a Uniform Distribution

26.1 Example

Let $X \sim \text{Uniform}(0, 1)$ and define

$$h(x) = \begin{cases} 0, & x \leq \frac{1}{2}, \\ 1, & x > \frac{1}{2}. \end{cases}$$

Let

$$Y = h(X).$$

Then

$$Y \sim \text{Bern}(p), \quad p = P(Y = 1) = P(X > 1/2) = 1/2.$$

27 Percentiles

27.1 Definition

Let X be a continuous random variable with cdf $F(x)$.

The 100 p th percentile of the distribution of X is the value x_p that satisfies:

$$F(x_p) = P(X \leq x_p) = p.$$

27.2 Example: Uniform Distribution

Let $X \sim \text{Uniform}(A, B)$ with

$$F(x) = \begin{cases} 0, & x < A, \\ \frac{x - A}{B - A}, & A \leq x < B, \\ 1, & x \geq B. \end{cases}$$

73rd Percentile

Solve:

$$F(x) = 0.73.$$

Clearly $A < x < B$, hence:

$$\begin{aligned} \frac{x - A}{B - A} &= 0.73 \\ \Rightarrow x &= A + 0.73(B - A). \end{aligned}$$

27.3 Example: Exponential Distribution

Let $X \sim \exp(\lambda)$ with cdf

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x > 0. \end{cases}$$

90th Percentile

Solve:

$$1 - e^{-\lambda x} = 0.9$$

$$e^{-\lambda x} = 0.1$$

$$-\lambda x = \ln(0.1)$$

$$x = -\frac{1}{\lambda} \ln(0.1).$$

27.4 Example: Piecewise Density

Given cdf:

$$F(x) = \begin{cases} 0, & x < -1, \\ \frac{1}{2} - \frac{x^2}{2}, & -1 \leq x < 0, \\ \frac{1}{2} + \frac{x^2}{2}, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

35th Percentile

Since

$$0.35 < F(0) = 0.5,$$

the percentile lies in $(-1, 0)$.

Solve:

$$\frac{1}{2} - \frac{x^2}{2} = 0.35$$

$$x^2 - 1 + 0.7 = 0$$

$$x^2 = 0.3$$

$$x = -\sqrt{0.3} \quad (\text{since } x \in (-1, 0)).$$

28 Memoryless Property

28.1 Exponential Distribution

Let $X \sim \exp(\lambda)$, $\lambda > 0$.

$$P(X > t) = e^{-\lambda t}, \quad t > 0.$$

Check:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

Thus, the exponential distribution satisfies the **memoryless property**.

29 Cauchy Distribution via Transformation

29.1 Example

Let

$$X \sim U\left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad Y = \tan X.$$

Then Y takes values in $(-\infty, \infty)$.

CDF of Y

$$F_Y(y) = P(Y \leq y) = P(X \leq \tan^{-1}(y)).$$

Using cdf of Uniform distribution:

$$F_Y(y) = \frac{\tan^{-1}(y) + \frac{\pi}{2}}{\pi} = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(y).$$

PDF of Y

Differentiate:

$$f_Y(y) = \frac{1}{\pi} \cdot \frac{1}{1 + y^2}, \quad -\infty < y < \infty.$$

This is the **standard Cauchy distribution**.

30 Expectation and Moments

Let X be a continuous r.v. with pdf $f(x)$. Then

$$E[h(X)] = \int_{-\infty}^{\infty} h(x)f(x) dx.$$

30.1 Mean

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx.$$

30.2 nth Moment

$$E(X^n) = \int_{-\infty}^{\infty} x^n f(x) dx.$$

30.3 Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

Important Note

A moment exists only if the integral is finite.

31 Examples of Moments

31.1 Uniform Distribution

Let $X \sim U(A, B)$,

$$f(x) = \frac{1}{B - A}, \quad A < x < B.$$

$$E(X) = \frac{A + B}{2}, \quad E(X^2) = \frac{A^2 + AB + B^2}{3},$$

$$\text{Var}(X) = \frac{(B - A)^2}{12}.$$

31.2 Exponential Distribution

Let $X \sim \exp(\lambda)$.

$$E(X) = \frac{1}{\lambda}, \quad E(X^2) = \frac{2}{\lambda^2}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

31.3 Non-example: Cauchy

If X is Cauchy,

$E(X)$ does not exist.

32 Moment Generating Function (MGF)

The mgf is

$$M_X(t) = E(e^{tX}).$$

32.1 Examples

Uniform

$$X \sim U(A, B) : \quad M_X(t) = \frac{e^{tB} - e^{tA}}{t(B - A)}.$$

Exponential

$$X \sim \exp(\lambda) : \quad M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

33 Normal Distribution

33.1 Definition

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

A random variable with this pdf is called

$$X \sim N(\mu, \sigma^2).$$

33.2 Standardization

Let

$$Z = \frac{X - \mu}{\sigma}.$$

Then

$$Z \sim N(0, 1), \quad f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

34 Normal Distribution (continued)

34.1 Definition and Basic Properties

The pdf of a normal distribution with parameters μ and σ^2 is

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

For $\mu = 0$, $\sigma^2 = 1$ we get

$$f(x; 0, 1) = \phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty.$$

The function $f(x; \mu, \sigma^2)$ is a pdf for every $-\infty < \mu < \infty$, $\sigma^2 > 0$, since it satisfies:

- $f(x; \mu, \sigma^2) \geq 0$ for all $x \in \mathbb{R}$,
- $\int_{-\infty}^{\infty} f(x; \mu, \sigma^2) dx = 1$.

34.2 Normal Random Variable

Defn. Let X be a r.v. with pdf $f(x; \mu, \sigma^2)$ as above. Then X is called a normal random variable with parameter μ and σ^2 , written

$$X \sim N(\mu, \sigma^2), \quad -\infty < \mu < \infty, \sigma^2 > 0.$$

In particular, if

$$X \sim N(0, 1),$$

then X is called a **standard normal** r.v.

34.3 Standardization

Proposition. Let $X \sim N(\mu, \sigma^2)$. Define

$$Z = \frac{X - \mu}{\sigma}.$$

Then

$$Z \sim N(0, 1).$$

Conversely, if $Z \sim N(0, 1)$, then

$$X = \mu + \sigma Z \sim N(\mu, \sigma^2).$$

Proof. Exercise.

34.4 Normalization Constant via Double Integral

To show

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma^2) dx = 1$$

it suffices to show

$$\int_{-\infty}^{\infty} \phi(z) dz = 1,$$

where we make the transformation

$$z = \frac{x - \mu}{\sigma} \quad \text{or} \quad x = \sigma z + \mu.$$

Let

$$C = \int_{-\infty}^{\infty} e^{-x^2/2} dx.$$

Then

$$C^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2/2} dy \right) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy.$$

Put

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Then

$$r^2 = x^2 + y^2, \quad \theta = \tan^{-1}(y/x),$$

with $0 \leq r < \infty$, $\theta \in [-\pi, \pi]$ and

$$dx dy = r dr d\theta.$$

The Jacobian matrix is

$$J = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix}, \quad |J| = |\det J| = r.$$

Hence

$$C^2 = \int_{-\pi}^{\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = 2\pi \int_0^{\infty} r e^{-r^2/2} dr.$$

Substitute $u = \frac{r^2}{2}$, $du = r dr$:

$$2\pi \int_0^{\infty} e^{-u} du = 2\pi.$$

Thus

$$C^2 = 2\pi, \quad C = \sqrt{2\pi}.$$

Hence

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz = \int_{-\infty}^{\infty} \phi(z) dz = 1.$$

Therefore

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma^2) dx = 1.$$

34.5 Normal Probabilities via Standardization

Let $a \leq b$ and $X \sim N(\mu, \sigma^2)$. Then

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

Using $Z = \frac{X - \mu}{\sigma}$, we have

$$P(a < X \leq b) = P\left(\frac{a - \mu}{\sigma} < Z \leq \frac{b - \mu}{\sigma}\right) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right),$$

where $Z \sim N(0, 1)$ and $\Phi(z) = P(Z \leq z)$ is the cdf of the standard normal r.v.

Example

$$P(\mu \leq X \leq \mu + 2\sigma) = \int_{\mu}^{\mu+2\sigma} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} dx.$$

With $z = \frac{x - \mu}{\sigma}$, $dx = \sigma dz$, this becomes

$$\int_0^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \Phi(2) - \Phi(0).$$

34.6 Standard Normal Probability Table

Standard Normal Probability Table (Standard Normal Cdf).

The tables can be used to find the different percentiles of Z :

- Median of Z (50th percentile): $z = 0$,
- 90th percentile: $z = 1.28$,
- 95th percentile: $z = 1.645$.

34.7 Percentiles of a General Normal

Question. Find the 90th percentile of X , where $X \sim N(\mu, \sigma^2)$.

The required percentile $x_{0.9}$ is such that

$$P(X \leq x_{0.9}) = F_X(x_{0.9}) = 0.9.$$

Using standardization:

$$P\left(\frac{X - \mu}{\sigma} \leq \frac{x_{0.9} - \mu}{\sigma}\right) = \Phi\left(\frac{x_{0.9} - \mu}{\sigma}\right) = 0.9.$$

We know $\Phi(1.28) = 0.9$, hence

$$\frac{x_{0.9} - \mu}{\sigma} = 1.28 \quad \Rightarrow \quad x_{0.9} = \mu + 1.28\sigma.$$

Similarly, for any $0 < p < 1$, if z_p is the p th percentile of Z , then

$$x_p = \mu + z_p\sigma.$$

34.8 Example: 10th Percentile of Z

Example. 10th percentile of Z .

For any $0 < p < 1$,

$$z_p = -z_{1-p} \quad (\text{by symmetry}).$$

Hence

$$z_{0.1} = -z_{0.9} = -1.28.$$

34.9 MGF, Mean and Variance of Normal

Remark. $E(X) = \mu$ and $V(X) = \sigma^2$ whenever $X \sim N(\mu, \sigma^2)$.

The mgf is

$$M_X(t) = E(e^{tX}), \quad -\infty < t < \infty.$$

For $X \sim N(\mu, \sigma^2)$,

$$M_X(t) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{tx} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

Combine exponents:

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} + tx\right) dx.$$

Rewrite the exponent and complete the square (steps as in notes):

$$-\frac{(x-\mu)^2}{2\sigma^2} + tx = -\frac{1}{2\sigma^2} \left[(x-\mu)^2 - 2\sigma^2 tx \right] = -\frac{1}{2\sigma^2} \left[(x-\mu-\sigma^2 t)^2 - (\sigma^2 t^2 + 2\mu t \sigma^2) \right].$$

Hence

$$M_X(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{\mu t + \frac{1}{2}\sigma^2 t^2} \int_{-\infty}^{\infty} \exp\left(-\frac{(x - \mu - \sigma^2 t)^2}{2\sigma^2}\right) dx.$$

The integral is 1 (it is the integral of a normal pdf), so finally

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

(Here we have used the fact that the integral of a normal pdf over $(-\infty, \infty)$ is 1.)

$$\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - (\mu + t\sigma^2))^2}{2\sigma^2}\right)$$

is the pdf of $N(\mu + t\sigma^2, \sigma^2)$.

34.10 Symmetry and Linear Transformations of Normal

Let $X \sim N(\mu, \sigma^2)$ with

$$f(x) = f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

The normal density is symmetric about μ :

$$f(x - \mu) = f(-(x - \mu)).$$

With

$$Z = \frac{X - \mu}{\sigma},$$

we have

$$Z \sim N(0, 1).$$

Example: Linear Transformation

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, find the distribution of Y .

The mgf of Y is

$$M_Y(t) = E(e^{tY}) = E(e^{t(aX+b)}) = e^{bt} M_X(at) = \exp\left(bt + a\mu t + \frac{1}{2}a^2\sigma^2 t^2\right), \quad -\infty < t < \infty.$$

Thus Y is normal with parameters

$$Y \sim N(a\mu + b, a^2\sigma^2).$$

34.11 Mean and Variance from the MGF

The moment of $X \sim N(\mu, \sigma^2)$ is given by

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right).$$

Differentiate:

$$M'_X(t) = (\mu + \sigma^2 t) \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right),$$

so

$$M'_X(0) = \mu \quad \Rightarrow \quad E(X) = \mu \quad (\text{mean of normal distribution}).$$

Again,

$$M''_X(t) = (\sigma^2 + (\mu + \sigma^2 t)^2) \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right),$$

hence

$$M''_X(0) = \sigma^2 + \mu^2, \quad E(X^2) = \mu^2 + \sigma^2.$$

Therefore

$$V(X) = E(X^2) - [E(X)]^2 = (\mu^2 + \sigma^2) - \mu^2 = \sigma^2.$$

34.12 Exercises (Normal Moments)

- (1) If $X \sim N(\mu, \sigma^2)$, find the 3rd and 4th moments of X .
- (2) Evaluate $E(X)$ and $E(X^2)$ using the pdf, i.e. evaluate

$$\int_{-\infty}^{\infty} x f(x) dx \quad \text{and} \quad \int_{-\infty}^{\infty} x^2 f(x) dx.$$

35 Gamma Distribution

35.1 Gamma Function

Defn. For $\alpha > 0$, the gamma function $\Gamma(\alpha)$ is defined by

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx.$$

35.2 Properties of $\Gamma(\alpha)$

1. For any $\alpha > 1$,

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1),$$

or equivalently

$$\Gamma(\alpha + 1) = \alpha \Gamma(\alpha).$$

2. For any positive integer n ,

$$\Gamma(n) = (n - 1)!.$$

3. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

35.3 Gamma Distribution: Definition

Defn. A continuous r.v. X is said to have a gamma distribution with parameter $\beta > 0$ and $\alpha > 0$ if its pdf is

$$f(x; \alpha, \beta) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, \quad x > 0,$$

and $f(x; \alpha, \beta) = 0$ otherwise.

Equivalently, with $\lambda = 1/\beta$,

$$f(x; \alpha, \beta) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0.$$

Let

$$f(x; \alpha) = f(x; \alpha, 1) = \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}, \quad x > 0,$$

and $f(x; \alpha) = 0$ otherwise.

Then

$$\int_0^\infty f(x; \alpha) dx = 1.$$

Exercise. Check that for $\alpha > 0$, $\beta > 0$,

$$\int_0^\infty f(x; \alpha, \beta) dx = 1.$$

35.4 Special Case: Exponential as Gamma

If $\alpha = 1$ and $\beta = 1/\lambda$, then

$$f(x; 1, \beta) = f(x; 1, 1/\lambda) = \lambda e^{-\lambda x}, \quad x > 0,$$

and 0 otherwise; i.e. $f(x; 1, 1/\lambda)$ is the pdf of $\text{Exp}(\lambda)$.

35.5 MGF and Moments of Gamma

Exercise. Find $M_X(t)$, the mgf for a $\text{Gamma}(\alpha, \beta)$ r.v. (with pdf as above).

The (known) results are

$$E(X) = \alpha\beta, \quad V(X) = \alpha\beta^2.$$

36 Jointly Distributed Continuous Random Variables

36.1 Definition

Let X, Y be continuous random variables. A joint probability density function (pdf) $f(x, y)$ for these variables is a function satisfying

$$\begin{aligned} f(x, y) &\geq 0, \quad \forall x, y, \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= 1. \end{aligned}$$

For any real numbers $a < b$, $c < d$,

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f(x, y) dx dy.$$

More generally, for a suitable set $A \subset \mathbb{R}^2$,

$$P((X, Y) \in A) = \iint_A f(x, y) dx dy.$$

36.2 Example: Busy Counters

A fast food restaurant has two counters: one drive-through and one inside. Let

X = proportion of time the drive-through counter is busy, Y = proportion of time the inside counter

Since these are proportions,

$$0 \leq X \leq 1, \quad 0 \leq Y \leq 1.$$

So the range (support) is

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}.$$

The joint pdf is

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2), & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Check that f is a pdf. We compute

$$\int_0^1 \int_0^1 \frac{6}{5}(x + y^2) dx dy.$$

First integrate with respect to x :

$$\int_0^1 (x + y^2) dx = \left[\frac{x^2}{2} + xy^2 \right]_0^1 = \frac{1}{2} + y^2.$$

Hence

$$\int_0^1 \int_0^1 \frac{6}{5}(x + y^2) dx dy = \frac{6}{5} \int_0^1 \left(\frac{1}{2} + y^2 \right) dy = \frac{6}{5} \left[\frac{y}{2} + \frac{y^3}{3} \right]_0^1 = \frac{6}{5} \cdot \frac{5}{6} = 1.$$

Thus f is a valid joint pdf.

Probability that neither counter is busy for more than one quarter of the time.

We want

$$P(0 \leq X \leq \frac{1}{4}, 0 \leq Y \leq \frac{1}{4}) = \int_0^{1/4} \int_0^{1/4} \frac{6}{5}(x + y^2) dx dy.$$

Inner integral:

$$\int_0^{1/4} (x + y^2) dx = \left[\frac{x^2}{2} + xy^2 \right]_0^{1/4} = \frac{1}{32} + \frac{y^2}{4}.$$

So

$$P = \frac{6}{5} \int_0^{1/4} \left(\frac{1}{32} + \frac{y^2}{4} \right) dy = \frac{6}{5} \left[\frac{y}{32} + \frac{y^3}{12} \right]_0^{1/4} = \frac{6}{5} \left(\frac{1}{128} + \frac{1}{768} \right) = \frac{7}{640} \approx 0.0109.$$

Probability that $0 \leq X \leq \frac{1}{4}$.

$$P(0 \leq X \leq \frac{1}{4}) = \int_0^{1/4} \int_0^1 \frac{6}{5}(x + y^2) dy dx.$$

Inner integral:

$$\int_0^1 (x + y^2) dy = x + \frac{1}{3},$$

so

$$P = \frac{6}{5} \int_0^{1/4} \left(x + \frac{1}{3} \right) dx = \frac{6}{5} \left[\frac{x^2}{2} + \frac{x}{3} \right]_0^{1/4} = \frac{6}{5} \left(\frac{1}{32} + \frac{1}{12} \right) = \frac{11}{80} \approx 0.1375.$$

36.3 Marginal Densities

Definition. For a joint pdf $f(x, y)$, the marginal pdfs of X and Y are

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

Example (continued). For $f(x, y) = \frac{6}{5}(x + y^2)$ on D ,

$$f_X(x) = \int_0^1 \frac{6}{5}(x + y^2) dy = \frac{6}{5} \left(x + \frac{1}{3} \right), \quad 0 \leq x \leq 1,$$

and $f_X(x) = 0$ otherwise.

Similarly,

$$f_Y(y) = \int_0^1 \frac{6}{5}(x + y^2) dx = \frac{6}{5} \left(\frac{1}{2} + y^2 \right), \quad 0 \leq y \leq 1,$$

and $f_Y(y) = 0$ otherwise.

36.4 Conditional Densities

Definition. If $f_Y(y) > 0$, the conditional pdf of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}, \quad -\infty < x < \infty.$$

If $f_X(x) > 0$, the conditional pdf of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}, \quad -\infty < y < \infty.$$

Example (continued).

$$f_{X|Y}(x|y) = \frac{\frac{6}{5}(x + y^2)}{\frac{6}{5}(\frac{1}{2} + y^2)} = \frac{x + y^2}{\frac{1}{2} + y^2}, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

For $y = 0.5$,

$$f_{X|Y}(x|0.5) = \frac{x + 0.25}{0.75} = \frac{4x + 1}{3}, \quad 0 \leq x \leq 1.$$

Exercise. Evaluate $P(0.1 \leq Y \leq 0.3 \mid X > 0.2)$.

Solution.

$$P(0.1 \leq Y \leq 0.3 \mid X > 0.2) = \frac{P(0.1 \leq Y \leq 0.3, X > 0.2)}{P(X > 0.2)}.$$

As found above,

$$P(X \leq 0.2) = \frac{13}{125}, \quad P(X > 0.2) = \frac{112}{125}.$$

Also,

$$P(0.1 \leq Y \leq 0.3, X > 0.2) = \int_{0.2}^1 \int_{0.1}^{0.3} \frac{6}{5}(x + y^2) dy dx = \frac{386}{3125}.$$

Therefore

$$P(0.1 \leq Y \leq 0.3 \mid X > 0.2) = \frac{386/3125}{112/125} = \frac{193}{1400} \approx 0.1379.$$

36.5 Independence

Definition. Continuous r.v.s X and Y are independent if

$$f(x, y) = f_X(x)f_Y(y) \quad \text{for all } x, y.$$

For more than two variables $X_1, \dots, X_n$ with joint pdf $f(x_1, \dots, x_n)$, we have for all $a_i < b_i$,

$$P(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \cdots dx_1.$$

36.6 Example: Minimum of i.i.d. Exponentials

Let $X_1, \dots, X_n$ denote lifetimes of n identical components, operating independently, with each $X_i \sim \exp(\lambda)$. Then $X_1, \dots, X_n$ are i.i.d. and the joint pdf is

$$f(x_1, \dots, x_n) = \prod_{i=1}^n \lambda e^{-\lambda x_i}, \quad x_i > 0.$$

The probability that all components are working at time t is

$$P(X_1 > t, \dots, X_n > t) = P\left(\min_{1 \leq i \leq n} X_i > t\right) = \prod_{i=1}^n P(X_i > t) = (e^{-\lambda t})^n = e^{-n\lambda t}.$$

Thus, for $Y = \min_{1 \leq i \leq n} X_i$,

$$P(Y > t) = e^{-n\lambda t}, \quad t > 0,$$

so $Y \sim \exp(n\lambda)$.

36.7 Another Joint pdf Example

Consider

$$f(x, y) = \begin{cases} 24xy, & 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Check that f is a pdf. Clearly $f(x, y) \geq 0$ on its support. We check the integral:

$$\int_0^1 \int_0^{1-x} 24xy \, dy \, dx.$$

Inner integral:

$$\int_0^{1-x} 24xy \, dy = 24x \left[\frac{y^2}{2} \right]_0^{1-x} = 12x(1-x)^2.$$

Thus

$$\int_0^1 12x(1-x)^2 \, dx = 12 \int_0^1 (x - 2x^2 + x^3) \, dx = 12 \left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right]_0^1 = 12 \cdot \frac{1}{12} = 1.$$

So f is a valid joint pdf.

Exercise. For the joint pdf in the previous example, find

$$P(X + Y \leq 0.5), \quad f_X(x), \quad f_Y(y), \quad f_{Y|X}(y | x), \quad f_{X|Y}(x | y),$$

and check whether X and Y are independent.

Definition. Let h be a function of two variables. Then the expectation of $h(X, Y)$, if it exists, is

$$E[h(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f_{X,Y}(x, y) \, dy \, dx,$$

where $f_{X,Y}$ is the joint pdf of (X, Y) .

Example. In the previous joint-pdf example, evaluate

$$E(X), E(Y), \text{Var}(X), \text{Var}(Y), \text{Cov}(X, Y), \rho(X, Y).$$

Solution outline. We use:

$$E(X) = \iint x f_{X,Y}(x, y) dy dx, \quad E(Y) = \iint y f_{X,Y}(x, y) dx dy.$$

Then

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2, & \text{Var}(Y) &= E(Y^2) - [E(Y)]^2, \\ \text{Cov}(X, Y) &= E(XY) - E(X)E(Y), & \rho(X, Y) &= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}. \end{aligned}$$

All these are computed by integrating x^2 , y^2 , or xy against the joint pdf on the region where it is positive.

Example (mixed nuts box). A 1 kg box of mixed nuts contains almonds, cashews, and peanuts.

- X = weight of almonds (kg) in a randomly chosen box.
- Y = weight of cashews (kg).
- Z = weight of peanuts (kg).

Then

$$X + Y + Z = 1, \quad 0 \leq X \leq 1, \quad 0 \leq Y \leq 1, \quad 0 \leq Z \leq 1.$$

Suppose the costs per kg are:

$$\text{Almonds: 250, Cashews: 200, Peanuts: 300.}$$

Let $h(x, y)$ be the cost of a box with x kg of almonds, y kg of cashews and $(1 - x - y)$ kg of peanuts:

$$h(x, y) = 250x + 200y + 300(1 - x - y) = 300 - 50x - 100y.$$

The average cost of the box is

$$E[h(X, Y)] = E[300 - 50X - 100Y] = 300 - 50E[X] - 100E[Y],$$

so once the joint pdf of (X, Y) (or their marginals) is known, $E[X]$ and $E[Y]$ can be computed and substituted in the above expression.

Example (sum of two exponentials). Let $X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$. Define

$$Y = X_1 + X_2.$$

We find the pdf of Y .

Note that Y takes values in $(0, \infty)$. For $y > 0$,

$$P(Y \leq y) = P(X_1 + X_2 \leq y) = \iint_{x_1 + x_2 \leq y} \lambda^2 e^{-\lambda(x_1 + x_2)} dx_1 dx_2.$$

One convenient way is to integrate first with respect to x_2 :

$$P(Y \leq y) = \int_0^y \left(\int_0^{y-x_1} \lambda^2 e^{-\lambda(x_1 + x_2)} dx_2 \right) dx_1.$$

Carrying out the inner integral and then the outer one gives

$$P(Y \leq y) = 1 - e^{-\lambda y} (1 + \lambda y), \quad y > 0.$$

Differentiating with respect to y , we obtain the pdf

$$f_Y(y) = \frac{d}{dy} P(Y \leq y) = \lambda^2 y e^{-\lambda y}, \quad y > 0,$$

and $f_Y(y) = 0$ for $y \leq 0$.

Thus

$$Y \sim \Gamma(2, 1/\lambda).$$

General transformation $Z = g(X, Y)$.

Suppose X, Y are continuous r.v.'s with joint pdf $f_{X,Y}(x, y)$ and define

$$Z = g(X, Y).$$

Then the cdf of Z is

$$F_Z(z) = P(Z \leq z) = P(g(X, Y) \leq z) = \iint_{g(x,y) \leq z} f_{X,Y}(x, y) dx dy.$$

Differentiating (where allowed) gives

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

as the pdf of Z .

Special case: $Z = X + Y$.

Then

$$F_Z(z) = P(X + Y \leq z) = \iint_{x+y \leq z} f_{X,Y}(x, y) dx dy.$$

Writing the integral explicitly with respect to y for fixed x :

$$F_Z(z) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{z-x} f_{X,Y}(x, y) dy \right) dx.$$

If we differentiate under the integral sign, we get

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z-x) dx.$$

Note. If X and Y are independent, then

$$f_{X,Y}(x, y) = f_X(x)f_Y(y),$$

so

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx.$$

The right-hand side is called the *convolution* of f_X and f_Y , denoted $f_X * f_Y$.

Remark.

(1) If X and Y are independent and both non-negative, then

$$f_Z(z) = \int_0^z f_X(x)f_Y(z-x) dx, \quad z > 0.$$

(2) The convolution of two pdfs is also a pdf (non-negative and integrates to 1).

Example. Let X, Y be iid $U(0, 1)$ and $Z = X + Y$.

$$f_X(x) = f_Y(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The range of $Z = X + Y$ is $(0, 2)$.

$$f_Z(z) = 0 \quad \text{if } z \leq 0 \text{ or } z \geq 2.$$

For $0 < z < 1$,

$$f_Z(z) = \int_0^z 1 \cdot 1 dx = z.$$

For $1 \leq z < 2$,

$$f_Z(z) = \int_{z-1}^1 1 \cdot 1 dx = 2 - z.$$

Thus the density of $Z = X + Y$ is

$$f_Z(z) = \begin{cases} z, & 0 < z < 1, \\ 2 - z, & 1 \leq z < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Graphically this is a triangle (hence sometimes called the *triangular distribution*).

Theorem. Let X and Y be independent random variables with

$$X \sim \Gamma(\alpha_1, \beta), \quad Y \sim \Gamma(\alpha_2, \beta).$$

Then

$$Z = X + Y \sim \Gamma(\alpha_1 + \alpha_2, \beta).$$

Proof. For $z \leq 0$, $f_Z(z) = 0$. For $z > 0$ we use the convolution formula:

$$f_Z(z) = \int_0^z f_X(x) f_Y(z-x) dx.$$

Substituting the gamma pdfs,

$$f_X(x) = \frac{x^{\alpha_1-1} e^{-x/\beta}}{\beta^{\alpha_1} \Gamma(\alpha_1)}, \quad f_Y(y) = \frac{y^{\alpha_2-1} e^{-y/\beta}}{\beta^{\alpha_2} \Gamma(\alpha_2)},$$

we get

$$\begin{aligned} f_Z(z) &= \int_0^z \frac{x^{\alpha_1-1} e^{-x/\beta}}{\beta^{\alpha_1} \Gamma(\alpha_1)} \cdot \frac{(z-x)^{\alpha_2-1} e^{-(z-x)/\beta}}{\beta^{\alpha_2} \Gamma(\alpha_2)} dx \\ &= \frac{e^{-z/\beta}}{\beta^{\alpha_1+\alpha_2} \Gamma(\alpha_1) \Gamma(\alpha_2)} \int_0^z x^{\alpha_1-1} (z-x)^{\alpha_2-1} dx. \end{aligned}$$

Now substitute $x = zu$; then $dx = z du$ and as x goes from 0 to z , u goes from 0 to 1.

Thus

$$\begin{aligned} f_Z(z) &= \frac{e^{-z/\beta}}{\beta^{\alpha_1+\alpha_2} \Gamma(\alpha_1) \Gamma(\alpha_2)} \int_0^1 (zu)^{\alpha_1-1} (z(1-u))^{\alpha_2-1} z du \\ &= \frac{e^{-z/\beta} z^{\alpha_1+\alpha_2-1}}{\beta^{\alpha_1+\alpha_2} \Gamma(\alpha_1) \Gamma(\alpha_2)} \int_0^1 u^{\alpha_1-1} (1-u)^{\alpha_2-1} du. \end{aligned}$$

The integral is the Beta function:

$$\int_0^1 u^{\alpha_1-1} (1-u)^{\alpha_2-1} du = B(\alpha_1, \alpha_2) = \frac{\Gamma(\alpha_1) \Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2)}.$$

Therefore

$$f_Z(z) = \frac{e^{-z/\beta} z^{\alpha_1+\alpha_2-1}}{\beta^{\alpha_1+\alpha_2} \Gamma(\alpha_1 + \alpha_2)}, \quad z > 0.$$

This is exactly the pdf of $\Gamma(\alpha_1 + \alpha_2, \beta)$, proving the claim.

Remark. The identity

$$B(\alpha_1, \alpha_2) = \int_0^1 u^{\alpha_1-1} (1-u)^{\alpha_2-1} du = \frac{\Gamma(\alpha_1) \Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2)}$$

defines the Beta function and is often used in problems involving Gamma distributions and convolutions.

37 Beta Distribution and Ratios of Random Variables

37.1 Beta Density

[Beta Density] Let $\alpha_1 > 0$, $\alpha_2 > 0$. The *Beta density* with parameters α_1, α_2 is defined by

$$f_{\alpha_1, \alpha_2}(u) = \begin{cases} \frac{u^{\alpha_1-1}(1-u)^{\alpha_2-1}}{B(\alpha_1, \alpha_2)}, & 0 < u < 1, \\ 0, & \text{otherwise,} \end{cases}$$

where

$$B(\alpha_1, \alpha_2) = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2)}$$

is the Beta function.

A random variable whose pdf is of the above form is called a *Beta random variable* with parameters α_1, α_2 .

37.2 Exercises with Normal Random Variables

[Square of a Normal rv] Let $X \sim N(0, 1)$. Find the pdf of X^2 .

Solution. Set $Y = X^2$, so $Y \geq 0$. For $y > 0$, we have

$$P(Y \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx.$$

Differentiate w.r.t. y using the change-of-variable formula for monotone pieces, or more simply use the standard rule for transformations:

There are two roots $x = \pm\sqrt{y}$ with $\frac{dx}{dy} = \frac{1}{2\sqrt{y}}$, $-\frac{1}{2\sqrt{y}}$. Hence

$$f_Y(y) = f_X(\sqrt{y})\frac{1}{2\sqrt{y}} + f_X(-\sqrt{y})\frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}, \quad y > 0.$$

Thus

$$X^2 \sim \chi_1^2$$

(a chi-square distribution with 1 degree of freedom).

[Scaled Normal] Let $Y \sim N(0, \sigma^2)$. Find the pdf of Y^2 .

Solution. Write $Y = \sigma Z$ where $Z \sim N(0, 1)$. Then

$$Y^2 = \sigma^2 Z^2.$$

Since $Z^2 \sim \chi_1^2$, it follows that

$$\frac{Y^2}{\sigma^2} \sim \chi_1^2,$$

so

$$f_{Y^2}(y) = \frac{1}{\sigma^2} f_{\chi_1^2}\left(\frac{y}{\sigma^2}\right) = \frac{1}{\sqrt{2\pi} \sigma \sqrt{y}} \exp\left(-\frac{y}{2\sigma^2}\right), \quad y > 0.$$

[Sum of Squares of Normals] Let X and Y be independent with $X, Y \sim N(0, 1)$. Find the pdf of

$$W = X^2 + Y^2.$$

Solution. Since $X^2 \sim \chi_1^2$, $Y^2 \sim \chi_1^2$, and they are independent, we have

$$W = X^2 + Y^2 \sim \chi_2^2,$$

i.e. chi-square with 2 d.f. Its pdf is

$$f_W(w) = \frac{1}{2^1 \Gamma(1)} w^{1-1} e^{-w/2} = \frac{1}{2} e^{-w/2}, \quad w > 0.$$

(If instead $X, Y \sim N(0, \sigma^2)$, then $W/\sigma^2 \sim \chi_2^2$.)

[Ratios: Cauchy and F distributions] Let X and Y be independent with $X, Y \sim N(0, 1)$.

(a) Find the distribution of $Z_1 = \frac{Y}{X}$.

(b) Find the distribution of $Z_2 = \frac{Y^2}{X^2}$.

Solution.

(a) It is a standard fact that the ratio of two independent $N(0, 1)$ random variables has a Cauchy distribution with pdf

$$f_{Z_1}(z) = \frac{1}{\pi(1+z^2)}, \quad z \in \mathbb{R}.$$

(b) Note that $X^2 \sim \chi_1^2$, $Y^2 \sim \chi_1^2$ and they are independent. Thus

$$Z_2 = \frac{Y^2}{X^2} = \frac{(Y^2/1)}{(X^2/1)}$$

has an F -distribution with $(1, 1)$ degrees of freedom:

$$Z_2 \sim F_{1,1}.$$

Its pdf is

$$f_{Z_2}(z) = \frac{1}{\pi} z^{-1/2} (1+z)^{-1}, \quad z > 0.$$

37.3 Ratio of Two Continuous Random Variables

[Distribution of $Z = Y/X$] Let X and Y be continuous random variables with joint pdf $f_{X,Y}(x, y)$. Define

$$Z = \frac{Y}{X}.$$

We derive the pdf of Z .

We first write the cdf:

$$F_Z(z) = P(Z \leq z) = P\left(\frac{Y}{X} \leq z\right).$$

The inequality $\frac{Y}{X} \leq z$ is equivalent to

$$\{(x, y) : y/x \leq z\} = \{x > 0, y \leq zx\} \cup \{x < 0, y \geq zx\}.$$

Hence

$$F_Z(z) = \int_0^\infty \int_{-\infty}^{zx} f_{X,Y}(x, y) dy dx + \int_{-\infty}^0 \int_{zx}^\infty f_{X,Y}(x, y) dy dx.$$

Now differentiate w.r.t. z (Leibniz rule). For the first integral,

$$\frac{d}{dz} \int_{-\infty}^{zx} f_{X,Y}(x, y) dy = f_{X,Y}(x, zx) \cdot x, \quad (x > 0),$$

and for the second,

$$\frac{d}{dz} \int_{zx}^\infty f_{X,Y}(x, y) dy = -f_{X,Y}(x, zx) \cdot x, \quad (x < 0).$$

Thus the pdf $f_Z(z)$ is

$$\begin{aligned} f_Z(z) &= \int_0^\infty x f_{X,Y}(x, zx) dx - \int_{-\infty}^0 x f_{X,Y}(x, zx) dx \\ &= \int_{-\infty}^\infty |x| f_{X,Y}(x, zx) dx, \end{aligned}$$

for all $z \in \mathbb{R}$ such that the integral exists.

So the general formula is:

$$f_Z(z) = \int_{-\infty}^\infty |x| f_{X,Y}(x, zx) dx, \quad z \in \mathbb{R}.$$

37.4 Ratio of Two Independent Gamma Variables

Suppose X and Y are independent r.v.'s with joint pdf

$$f_{X,Y}(x, y) = f_X(x)f_Y(y).$$

Then the formula above simplifies to

$$f_Z(z) = \int_{-\infty}^\infty |x| f_X(x)f_Y(zx) dx.$$

Theorem: Beta-Prime (Inverted Beta) Distribution

Let $X \sim \Gamma(\alpha_1, \beta)$ and $Y \sim \Gamma(\alpha_2, \beta)$ be independent, where $\alpha_1 > 0$, $\alpha_2 > 0$, $\beta > 0$. Define

$$Z = \frac{X}{Y}.$$

Then Z has pdf

$$f_Z(z) = \begin{cases} \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} z^{\alpha_1-1} (1+z)^{-(\alpha_1+\alpha_2)}, & z > 0, \\ 0, & \text{otherwise,} \end{cases}$$

i.e. Z has a *Beta-prime* (or *inverted Beta*, or *Beta of the second kind*) distribution with parameters (α_1, α_2) .

Proof. The gamma pdfs are

$$f_X(x) = \begin{cases} \frac{1}{\beta^{\alpha_1}\Gamma(\alpha_1)} x^{\alpha_1-1} e^{-x/\beta}, & x > 0, \\ 0, & x \leq 0, \end{cases} \quad f_Y(y) = \begin{cases} \frac{1}{\beta^{\alpha_2}\Gamma(\alpha_2)} y^{\alpha_2-1} e^{-y/\beta}, & y > 0, \\ 0, & y \leq 0. \end{cases}$$

Since $X, Y > 0$ a.s., the integral reduces to $x > 0$:

$$f_Z(z) = \int_0^\infty x f_X(x) f_Y(zx) dx, \quad z > 0,$$

and $f_Z(z) = 0$ for $z \leq 0$.

Substitute:

$$\begin{aligned} f_Z(z) &= \int_0^\infty x \frac{x^{\alpha_1-1} e^{-x/\beta}}{\beta^{\alpha_1}\Gamma(\alpha_1)} \cdot \frac{(zx)^{\alpha_2-1} e^{-zx/\beta}}{\beta^{\alpha_2}\Gamma(\alpha_2)} dx \\ &= \frac{z^{\alpha_2-1}}{\beta^{\alpha_1+\alpha_2}\Gamma(\alpha_1)\Gamma(\alpha_2)} \int_0^\infty x^{\alpha_1+\alpha_2-1} e^{-(1+z)x/\beta} dx. \end{aligned}$$

Now set $t = \frac{(1+z)x}{\beta}$, so $x = \frac{\beta}{1+z}t$ and $dx = \frac{\beta}{1+z}dt$. Then

$$\begin{aligned} \int_0^\infty x^{\alpha_1+\alpha_2-1} e^{-(1+z)x/\beta} dx &= \left(\frac{\beta}{1+z}\right)^{\alpha_1+\alpha_2} \int_0^\infty t^{\alpha_1+\alpha_2-1} e^{-t} dt \\ &= \left(\frac{\beta}{1+z}\right)^{\alpha_1+\alpha_2} \Gamma(\alpha_1 + \alpha_2). \end{aligned}$$

Thus

$$\begin{aligned} f_Z(z) &= \frac{z^{\alpha_2-1}}{\beta^{\alpha_1+\alpha_2}\Gamma(\alpha_1)\Gamma(\alpha_2)} \left(\frac{\beta}{1+z}\right)^{\alpha_1+\alpha_2} \Gamma(\alpha_1 + \alpha_2) \\ &= \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} z^{\alpha_2-1} (1+z)^{-(\alpha_1+\alpha_2)}, \quad z > 0, \end{aligned}$$

which matches the claimed Beta-prime pdf (note that the roles α_1, α_2 may be interchanged depending on whether $Z = X/Y$ or $Z = Y/X$). This completes the proof. $\square$

The Beta-prime density integrates to 1 because

$$\int_0^\infty z^{\alpha_1-1}(1+z)^{-(\alpha_1+\alpha_2)} dz = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2)} = B(\alpha_1, \alpha_2).$$

Hence the normalising constant $1/B(\alpha_1, \alpha_2)$ appears naturally.

37.5 Recall: Beta Random Variable

A Beta random variable X with parameters $\alpha_1 > 0$, $\alpha_2 > 0$ has pdf

$$f_X(x) = \begin{cases} \frac{1}{B(\alpha_1, \alpha_2)} x^{\alpha_1-1} (1-x)^{\alpha_2-1}, & 0 < x < 1, \\ 0, & \text{otherwise,} \end{cases}$$

where

$$B(\alpha_1, \alpha_2) = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2)}.$$

This connects the Beta distribution (support in $(0, 1)$) with the Beta-prime / inverted Beta distribution (support in $(0, \infty)$) via transformations such as $Z = \frac{X}{1-X}$ and its inverse $X = \frac{Z}{1+Z}$.

38 Beta Distribution

Let $X \sim \text{Beta}(\alpha_1, \alpha_2)$ with pdf

$$f_X(x) = \frac{1}{B(\alpha_1, \alpha_2)} x^{\alpha_1-1} (1-x)^{\alpha_2-1}, \quad 0 < x < 1,$$

where $B(\alpha_1, \alpha_2)$ is the Beta function.

38.1 Mean

$$E[X] = \int_0^1 x f_X(x) dx = \frac{1}{B(\alpha_1, \alpha_2)} \int_0^1 x^{\alpha_1} (1-x)^{\alpha_2-1} dx = \frac{B(\alpha_1 + 1, \alpha_2)}{B(\alpha_1, \alpha_2)}.$$

$$\text{Using } B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

$$\begin{aligned} E[X] &= \frac{\Gamma(\alpha_1 + 1)\Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2 + 1)} \cdot \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} \\ &= \frac{\alpha_1 \Gamma(\alpha_1)\Gamma(\alpha_2)}{(\alpha_1 + \alpha_2)\Gamma(\alpha_1 + \alpha_2)} \cdot \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} \\ &= \frac{\alpha_1}{\alpha_1 + \alpha_2}. \end{aligned}$$

So

$$\boxed{E[X] = \frac{\alpha_1}{\alpha_1 + \alpha_2}}.$$

38.2 Second moment

Similarly,

$$E[X^2] = \int_0^1 x^2 f_X(x) dx = \frac{1}{B(\alpha_1, \alpha_2)} \int_0^1 x^{\alpha_1+1} (1-x)^{\alpha_2-1} dx = \frac{B(\alpha_1+2, \alpha_2)}{B(\alpha_1, \alpha_2)}.$$

Hence

$$\begin{aligned} E[X^2] &= \frac{\Gamma(\alpha_1+2)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2+2)} \cdot \frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} \\ &= \frac{\alpha_1(\alpha_1+1)}{(\alpha_1+\alpha_2)(\alpha_1+\alpha_2+1)}. \end{aligned}$$

Thus

$$E[X^2] = \frac{\alpha_1(\alpha_1+1)}{(\alpha_1+\alpha_2)(\alpha_1+\alpha_2+1)}.$$

38.3 Variance

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= \frac{\alpha_1(\alpha_1+1)}{(\alpha_1+\alpha_2)(\alpha_1+\alpha_2+1)} - \left(\frac{\alpha_1}{\alpha_1+\alpha_2} \right)^2 \\ &= \frac{\alpha_1(\alpha_1+1)(\alpha_1+\alpha_2) - \alpha_1^2(\alpha_1+\alpha_2+1)}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)} \\ &= \frac{\alpha_1[\alpha_1 + \alpha_1\alpha_2 + \alpha_2 - \alpha_1^2 - \alpha_1\alpha_2 - \alpha_1]}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)} \\ &= \frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}. \end{aligned}$$

So

$$\text{Var}(X) = \frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}.$$

39 Multivariate Random Variables and Linear Transformations

39.1 Linear transformation of a random vector

Let $X = (X_1, \dots, X_n)^\top$ be a continuous random vector with joint pdf

$$f_X(x_1, \dots, x_n) = f_X(\mathbf{x}).$$

Let $A = (a_{ij})$ be an $n \times n$ matrix with $\det A \neq 0$ and define

$$Y_i = \sum_{j=1}^n a_{ij} X_j, \quad i = 1, \dots, n.$$

In vector form,

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} = A \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix} = AX.$$

Theorem (Jacobian for linear transform)

In the above setup, the joint pdf of $Y = (Y_1, \dots, Y_n)$ is

$$f_Y(\mathbf{y}) = \frac{1}{|\det A|} f_X(A^{-1}\mathbf{y}),$$

because the Jacobian determinant of the transformation $x = A^{-1}y$ is $|\det A^{-1}| = 1/|\det A|$.

If we write $B = A^{-1} = (b_{ij})$, then

$$X_i = \sum_{j=1}^n b_{ij} Y_j, \quad i = 1, \dots, n,$$

and f_Y is obtained by substituting these X_i into f_X and multiplying by $1/|\det A|$.

40 Example: Partial Sums of Exponential Variables

Let $X_1, \dots, X_n$ be i.i.d. $\text{Exp}(\lambda)$, $\lambda > 0$. The joint pdf of X is

$$f_X(x_1, \dots, x_n) = \lambda^n \exp(-\lambda(x_1 + \dots + x_n)), \quad x_i > 0, \quad i = 1, \dots, n,$$

and 0 otherwise.

Define the partial sums

$$S_k = X_1 + X_2 + \dots + X_k, \quad 1 \leq k \leq n.$$

So

$$S_1 = X_1, \quad S_2 = X_1 + X_2, \quad \dots, \quad S_n = X_1 + \dots + X_n.$$

40.1 The transformation matrix

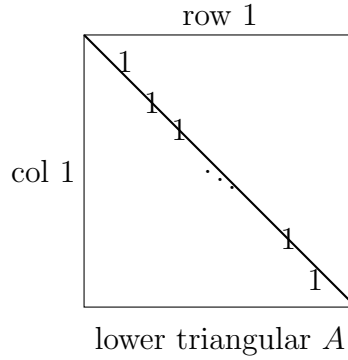
In vector form,

$$S = \begin{pmatrix} S_1 \\ S_2 \\ \vdots \\ S_n \end{pmatrix} = A \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix},$$

where A is the $n \times n$ lower-triangular matrix

$$A = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{pmatrix}.$$

Figure: schematic of A



Because A is lower triangular with all diagonal entries equal to 1, its determinant is

$$\det A = 1 \cdot 1 \cdots 1 = 1.$$

The inverse transformation is

$$X_1 = S_1, \quad X_k = S_k - S_{k-1}, \quad 2 \leq k \leq n.$$

40.2 Region of $(S_1, \dots, S_n)$

Since $X_i > 0$, we have

$$S_1 = X_1 > 0, \quad S_2 - S_1 = X_2 > 0, \quad \dots, \quad S_n - S_{n-1} = X_n > 0.$$

Thus the support of $S = (S_1, \dots, S_n)$ is

$$0 < S_1 < S_2 < \cdots < S_n < \infty.$$

40.3 Joint pdf of $(S_1, \dots, S_n)$

We use the Jacobian of the inverse map $x = A^{-1}s$. Since $\det A = 1$, we also have $\det A^{-1} = 1$. Hence

$$f_S(s_1, \dots, s_n) = f_X(x_1, \dots, x_n) \Big|_{x_1=s_1, x_k=s_k-s_{k-1}, k \geq 2}.$$

We get

$$\begin{aligned}
f_S(s_1, \dots, s_n) &= \lambda^n \exp \left(-\lambda \sum_{k=1}^n x_k \right) \\
&= \lambda^n \exp \left(-\lambda [s_1 + (s_2 - s_1) + \dots + (s_n - s_{n-1})] \right) \\
&= \lambda^n e^{-\lambda s_n}
\end{aligned}$$

for $0 < s_1 < s_2 < \dots < s_n < \infty$, and

$$f_S(s_1, \dots, s_n) = 0 \quad \text{otherwise.}$$

So the joint pdf is

$$f_{S_1, \dots, S_n}(s_1, \dots, s_n) = \begin{cases} \lambda^n e^{-\lambda s_n}, & 0 < s_1 < s_2 < \dots < s_n < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

41 Independence of Functions of Disjoint Subsets

Let $X_1, \dots, X_n$ be independent random variables. Let Y be any function of $(X_1, \dots, X_m)$ for some $m < n$, and let Z be any function of $(X_{m+1}, \dots, X_n)$. Then Y and Z are independent.

Idea of proof. Because $X_1, \dots, X_n$ are independent, the vector

$$(X_1, \dots, X_m) \quad \text{is independent of} \quad (X_{m+1}, \dots, X_n).$$

Any measurable function of the first block is independent of any measurable function of the second block. Therefore any $Y = g(X_1, \dots, X_m)$ is independent of any $Z = h(X_{m+1}, \dots, X_n)$.

42 Moment Generating Functions and Independence

42.1 Recap: Independence of Functions of Disjoint Subsets

Let $X_1, \dots, X_n$ be independent random variables. Let Y be any (measurable) function of $(X_1, \dots, X_m)$ for some $m < n$, and let Z be any function of $(X_{m+1}, \dots, X_n)$. Then Y and Z are independent.

42.2 Moment Generating Functions (mgf)

For a random variable X , the *moment generating function* (mgf) of X is

$$M_X(t) = E[e^{tX}],$$

defined for all real t for which the expectation (right-hand side) is finite.

[Joint mgf] Let (X, Y) be a pair of jointly distributed random variables. The *joint mgf* of (X, Y) is

$$M_{X,Y}(s, t) = E[e^{sX + tY}],$$

defined for all (s, t) for which the expectation is finite.

42.3 Uniqueness of mgf

[Uniqueness]

(a) Suppose X and Y are random variables such that

$$M_X(t) = M_Y(t) \quad \text{for all } t \in (-t_0, t_0)$$

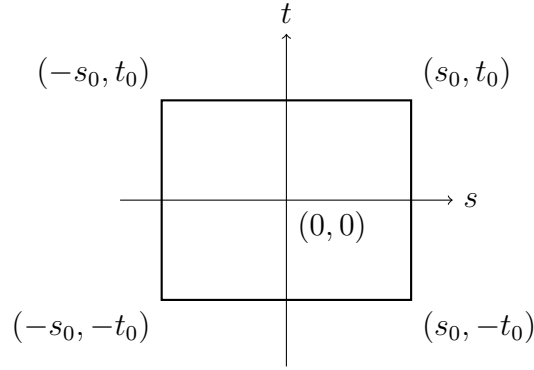
for some $t_0 > 0$. Then X and Y have the same distribution.

(b) Suppose (X_1, Y_1) and (X_2, Y_2) are pairs of random variables such that

$$M_{X_1, Y_1}(s, t) = M_{X_2, Y_2}(s, t) \quad \text{for all } (s, t) \text{ in some rectangle containing } (0, 0).$$

Then (X_1, Y_1) and (X_2, Y_2) have the same joint distribution.

Figure: rectangle around $(0, 0)$ in the (s, t) -plane



42.4 Independence and joint mgf

Let X and Y be random variables with joint mgf $M_{X,Y}(s, t)$. Then X and Y are independent if and only if

$$M_{X,Y}(s, t) = M_X(s) M_Y(t)$$

for all (s, t) in a rectangle $(-s_0, s_0) \times (-t_0, t_0)$ for some $s_0, t_0 > 0$.

Proof. ($\Rightarrow$) If X and Y are independent, then

$$\begin{aligned} M_{X,Y}(s, t) &= E[e^{sX+tY}] = E[e^{sX} e^{tY}] = E[e^{sX}] E[e^{tY}] \\ &= M_X(s) M_Y(t), \end{aligned}$$

for all (s, t) for which both mgfs are finite.

($\Leftarrow$) Conversely, suppose

$$M_{X,Y}(s, t) = M_X(s) M_Y(t) \quad \text{for all } (s, t) \in (-s_0, s_0) \times (-t_0, t_0). \quad (*)$$

Let $\hat{X}$ and $\hat{Y}$ be independent random variables such that $\hat{X} \stackrel{d}{=} X$ and $\hat{Y} \stackrel{d}{=} Y$, i.e.

$$M_{\hat{X}}(t) = M_X(t), \quad M_{\hat{Y}}(t) = M_Y(t) \quad \text{for } t \in (-t_0, t_0).$$

Since $\hat{X}$ and $\hat{Y}$ are independent, their joint mgf is

$$M_{\hat{X}, \hat{Y}}(s, t) = M_{\hat{X}}(s)M_{\hat{Y}}(t) = M_X(s)M_Y(t).$$

By (*),

$$M_{X,Y}(s, t) = M_{\hat{X}, \hat{Y}}(s, t)$$

on a rectangle containing $(0, 0)$. By the uniqueness theorem for joint mgfs, (X, Y) and $(\hat{X}, \hat{Y})$ have the same joint distribution. But $(\hat{X}, \hat{Y})$ are independent, so (X, Y) are also independent.

Equivalently, for all $a, b \in \mathbb{R}$,

$$P(X \leq a, Y \leq b) = P(\hat{X} \leq a, \hat{Y} \leq b) = P(\hat{X} \leq a) P(\hat{Y} \leq b) = P(X \leq a) P(Y \leq b),$$

which is exactly the definition of independence. $\square$

42.5 Example: Linear combinations of independent normals

[Two variables] Let $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ be independent. Let $a, b \in \mathbb{R}$ and consider $Z = aX + bY$.

The mgfs of X and Y are

$$M_X(t) = \exp(\mu_1 t + \tfrac{1}{2}\sigma_1^2 t^2), \quad M_Y(t) = \exp(\mu_2 t + \tfrac{1}{2}\sigma_2^2 t^2).$$

Since $Z = aX + bY$,

$$M_Z(t) = E[e^{t(aX+bY)}] = M_{X,Y}(at, bt) = M_X(at)M_Y(bt),$$

using independence and the previous theorem.

Hence

$$\begin{aligned} M_Z(t) &= \exp(\mu_1(at) + \tfrac{1}{2}\sigma_1^2(at)^2) \exp(\mu_2(bt) + \tfrac{1}{2}\sigma_2^2(bt)^2) \\ &= \exp((a\mu_1 + b\mu_2)t + \tfrac{1}{2}(a^2\sigma_1^2 + b^2\sigma_2^2)t^2). \end{aligned}$$

This is the mgf of a normal distribution with mean $a\mu_1 + b\mu_2$ and variance $a^2\sigma_1^2 + b^2\sigma_2^2$. Therefore

$$\boxed{Z = aX + bY \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2).}$$

[Linear combination of independent normals] Let $X_1, \dots, X_n$ be independent random variables with

$$X_i \sim N(\mu_i, \sigma_i^2), \quad i = 1, \dots, n,$$

and let $a_1, \dots, a_n \in \mathbb{R}$. Define

$$Y = \sum_{i=1}^n a_i X_i.$$

Then Y is normally distributed with

$$Y \sim N\left(\sum_{i=1}^n a_i \mu_i, \sum_{i=1}^n a_i^2 \sigma_i^2\right).$$

Proof. Using independence,

$$M_Y(t) = E[e^{t \sum a_i X_i}] = E\left[\prod_{i=1}^n e^{ta_i X_i}\right] = \prod_{i=1}^n E[e^{ta_i X_i}] = \prod_{i=1}^n M_{X_i}(a_i t).$$

Each X_i has mgf

$$M_{X_i}(t) = \exp\left(\mu_i t + \frac{1}{2}\sigma_i^2 t^2\right),$$

so

$$\begin{aligned} M_Y(t) &= \prod_{i=1}^n \exp\left(\mu_i(a_i t) + \frac{1}{2}\sigma_i^2(a_i t)^2\right) \\ &= \exp\left(\sum_{i=1}^n a_i \mu_i t + \frac{1}{2} \sum_{i=1}^n a_i^2 \sigma_i^2 t^2\right). \end{aligned}$$

This is the mgf of a normal distribution with mean $\sum_{i=1}^n a_i \mu_i$ and variance $\sum_{i=1}^n a_i^2 \sigma_i^2$, so the result follows. $\square$

43 Joint MGF and Independence

43.1 MGF of a Random Variable

For any random variable X , the moment generating function (mgf) is defined as

$$M_X(t) = E[e^{tX}],$$

for all t for which the expectation exists and is finite.

43.2 Joint MGF of (X, Y)

If X and Y are jointly distributed random variables, the joint mgf is defined as

$$M_{X,Y}(s, t) = E[e^{sX+tY}],$$

for all (s, t) such that the expectation is finite.

43.3 Uniqueness Properties of MGFs

43.3.1 Property (a)

Suppose X and Y are random variables such that

$$M_X(t) = M_Y(t) \quad \forall t \in (-t_0, t_0).$$

Then X and Y have the **same distribution**.

43.3.2 Property (b)

Suppose (X_1, X_2) and (Y_1, Y_2) satisfy

$$M_{X_1, X_2}(s, t) = M_{Y_1, Y_2}(s, t)$$

for all (s, t) in some rectangle containing $(0, 0)$. Then

$$(X_1, X_2) \stackrel{d}{=} (Y_1, Y_2).$$

43.4 Independence via MGFs

Let X and Y be random variables. Then X and Y are independent if and only if

$$M_{X,Y}(s, t) = M_X(s)M_Y(t)$$

for all (s, t) in some neighborhood of $(0, 0)$.

Proof

If X and Y are independent, then

$$M_{X,Y}(s, t) = E[e^{sX+tY}] = E[e^{sX}] E[e^{tY}] = M_X(s)M_Y(t).$$

Conversely, assume $M_{X,Y}(s, t) = M_X(s)M_Y(t)$. Take independent copies $\tilde{X}, \tilde{Y}$ such that

$$M_{\tilde{X}}(t) = M_X(t), \quad M_{\tilde{Y}}(t) = M_Y(t).$$

Then

$$M_{\tilde{X}, \tilde{Y}}(s, t) = M_X(s)M_Y(t) = M_{X,Y}(s, t).$$

By uniqueness of mgfs,

$$(X, Y) \stackrel{d}{=} (\tilde{X}, \tilde{Y})$$

and hence X and Y must be independent. □

43.5 Example: Linear Combination of Independent Normals

Suppose

$$X \sim N(\mu_1, \sigma_1^2), \quad Y \sim N(\mu_2, \sigma_2^2),$$

and $X \perp Y$. Let $a, b \in \mathbb{R}$. Consider

$$Z = aX + bY.$$

Then its mgf is

$$M_Z(t) = E[e^{t(aX+bY)}] = E[e^{taX}] E[e^{tbY}] = M_X(at)M_Y(bt).$$

Using the mgf of a normal,

$$M_X(at) = \exp\left(a\mu_1 t + \frac{1}{2}a^2\sigma_1^2 t^2\right),$$

$$M_Y(bt) = \exp\left(b\mu_2 t + \frac{1}{2}b^2\sigma_2^2 t^2\right).$$

Thus

$$Z \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2).$$

43.6 Sum of Independent Normals

If $X_1, \dots, X_n$ are independent with

$$X_i \sim N(\mu_i, \sigma_i^2),$$

and $a_i \in \mathbb{R}$, then

$$Y = \sum_{i=1}^n a_i X_i \sim N\left(\sum_{i=1}^n a_i \mu_i, \sum_{i=1}^n a_i^2 \sigma_i^2\right).$$

44 Bivariate Normal Distribution

A pair of random variables (X, Y) is called **bivariate normal** if

$$aX + bY \text{ is normally distributed for all } a, b \in \mathbb{R}.$$

44.1 Remarks

1. If (X, Y) is bivariate normal, then

$$X \sim N(\mu_X, \sigma_X^2), \quad Y \sim N(\mu_Y, \sigma_Y^2).$$

2. If (X, Y) is bivariate normal, then for constants c, d ,

$$cX + dY \text{ is also normal.}$$

3. If X and Y are each marginally normal, this does **not** imply bivariate normality.

44.2 Theorem

Suppose (X, Y) and (Z, W) are each bivariate normal and satisfy:

$$\begin{aligned} E[X] &= E[Z] = \mu_1, & E[Y] &= E[W] = \mu_2, \\ \text{Var}(X) &= \text{Var}(Z) = \sigma_1^2, & \text{Var}(Y) &= \text{Var}(W) = \sigma_2^2, \\ \text{Cov}(X, Y) &= \text{Cov}(Z, W) = \rho\sigma_1\sigma_2. \end{aligned}$$

Then

$$(X, Y) \stackrel{d}{=} (Z, W).$$

Proof Sketch

Take constants $s, t \in \mathbb{R}$. Consider the linear combinations

$$sX + tY, \quad sZ + tW.$$

Using the properties of normal distributions:

$$\begin{aligned} sX + tY &\sim N(s\mu_1 + t\mu_2, s^2\sigma_1^2 + t^2\sigma_2^2 + 2st\rho\sigma_1\sigma_2), \\ sZ + tW &\sim N(s\mu_1 + t\mu_2, s^2\sigma_1^2 + t^2\sigma_2^2 + 2st\rho\sigma_1\sigma_2). \end{aligned}$$

Hence their mgfs are equal, and by uniqueness,

$$(X, Y) \sim (Z, W).$$

45 Independence in Bivariate Normal

If (X, Y) is bivariate normal, then

$$X \text{ and } Y \text{ are independent} \iff \text{Cov}(X, Y) = 0.$$

Proof

If X and Y are independent, covariance is zero.

Conversely, assume $\text{Cov}(X, Y) = 0$. Then

$$M_{X,Y}(s, t) = E[e^{sX+tY}] = \exp\left(s\mu_1 + t\mu_2 + \frac{1}{2}(s^2\sigma_1^2 + t^2\sigma_2^2)\right) = M_X(s)M_Y(t).$$

Hence X and Y are independent. □

46 Bivariate Normal Distribution

We consider a pair of random variables (Y_1, Y_2) that follow a bivariate normal distribution. Individually,

$$Y_1 \sim N(\mu_1, \sigma_1^2),$$

$$Y_2 \sim N(\mu_2, \sigma_2^2),$$

and the covariance is given by

$$\text{Cov}(Y_1, Y_2) = \sigma_{12}.$$

The correlation coefficient is defined as

$$\rho = \frac{\sigma_{12}}{\sigma_1\sigma_2}, \quad |\rho| < 1.$$

Our task is to derive the joint pdf $g(y_1, y_2)$.

$$F(a, b) = P(Y_1 \leq a, Y_2 \leq b) = \int_{-\infty}^a \int_{-\infty}^b g(y_1, y_2) dy_2 dy_1. \quad (1)$$

47 Standardization of (Y_1, Y_2)

47.1 Introducing Standard Normals

We define

$$Z_1 = \frac{Y_1 - \mu_1}{\sigma_1}, \quad Z_2 = \frac{Y_2 - \mu_2}{\sigma_2}.$$

Then the standardized pair satisfies

$$(Z_1, Z_2) \sim \text{bivariate normal with } E(Z_1) = E(Z_2) = 0, \quad V(Z_1) = V(Z_2) = 1,$$

and covariance

$$\text{Cov}(Z_1, Z_2) = \rho.$$

Thus,

$$(Z_1, Z_2) \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right).$$

47.2 Constructing Equivalent Variables (U, V)

Your notes define:

$$\begin{aligned}U &= \sigma_1 Z_1 + \mu_1, \\V &= \sigma_2 \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right) + \mu_2.\end{aligned}$$

These are linear combinations of jointly normal variables, so (U, V) is also bivariate normal.

47.3 Checking Means

$$\begin{aligned}E(U) &= \sigma_1 E(Z_1) + \mu_1 = \mu_1, \\E(V) &= \sigma_2 \left(\rho E(Z_1) + \sqrt{1 - \rho^2} E(Z_2) \right) + \mu_2 = \mu_2.\end{aligned}$$

47.4 Checking Variances

$$\begin{aligned}V(U) &= \sigma_1^2 V(Z_1) = \sigma_1^2, \\V(V) &= \sigma_2^2 \left(\rho^2 V(Z_1) + (1 - \rho^2) V(Z_2) \right) = \sigma_2^2.\end{aligned}$$

47.5 Checking Covariance (your scratched work)

$$\begin{aligned}\text{Cov}(U, V) &= \text{Cov} \left(\sigma_1 Z_1 + \mu_1, \sigma_2 (\rho Z_1 + \sqrt{1 - \rho^2} Z_2) + \mu_2 \right) \\&= \sigma_1 \sigma_2 \left(\rho \text{Cov}(Z_1, Z_1) + \sqrt{1 - \rho^2} \text{Cov}(Z_1, Z_2) \right) \\&= \sigma_1 \sigma_2 (\rho + \sqrt{1 - \rho^2} \rho) \\&= \sigma_1 \sigma_2 \rho \\&= \sigma_{12}.\end{aligned}$$

Thus the constructed pair (U, V) has the same mean vector and covariance matrix as (Y_1, Y_2) . Therefore,

$$(U, V) \stackrel{d}{=} (Y_1, Y_2).$$

48 Probability Transformation

$$P(Y_1 \leq a, Y_2 \leq b) = P(U \leq a, V \leq b).$$

48.1 Transforming Bounds to (Z_1, Z_2)

From $U = \sigma_1 Z_1 + \mu_1$:

$$Z_1 = \frac{U - \mu_1}{\sigma_1} \Rightarrow z_1 \leq \frac{a - \mu_1}{\sigma_1}.$$

From

$$V = \sigma_2 \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right) + \mu_2,$$

we obtain:

$$Z_2 = \frac{V - \mu_2}{\sigma_2 \sqrt{1 - \rho^2}} - \frac{\rho Z_1}{\sqrt{1 - \rho^2}}.$$

Thus,

$$z_2 \leq \frac{b - \mu_2}{\sigma_2 \sqrt{1 - \rho^2}} - \frac{\rho z_1}{\sqrt{1 - \rho^2}}.$$

48.2 Writing the Double Integral

Using the standard bivariate normal joint density:

$$\frac{1}{2\pi} \exp \left(-\frac{z_1^2 + z_2^2}{2} \right),$$

we have:

$$P(Y_1 \leq a, Y_2 \leq b) = \int_{-\infty}^{(a - \mu_1)/\sigma_1} \int_{-\infty}^{\frac{b - \mu_2}{\sigma_2 \sqrt{1 - \rho^2}} - \frac{\rho z_1}{\sqrt{1 - \rho^2}}} \frac{1}{2\pi} \exp \left(-\frac{z_1^2 + z_2^2}{2} \right) dz_2 dz_1.$$

49 Manipulating the Exponent

Your notes expand the exponent inside the integral:

$$z_1 = \frac{y_1 - \mu_1}{\sigma_1}, \quad z_2 = \frac{y_2 - \mu_2}{\sigma_2}.$$

Then,

$$z_1^2 + z_2^2 - 2\rho z_1 z_2 = \frac{(y_1 - \mu_1)^2}{\sigma_1^2} + \frac{(y_2 - \mu_2)^2}{\sigma_2^2} - 2\rho \frac{(y_1 - \mu_1)(y_2 - \mu_2)}{\sigma_1 \sigma_2}.$$

This gives

$$\frac{1}{2(1 - \rho^2)} \left[\frac{(y_1 - \mu_1)^2}{\sigma_1^2} + \frac{(y_2 - \mu_2)^2}{\sigma_2^2} - 2\rho \frac{(y_1 - \mu_1)(y_2 - \mu_2)}{\sigma_1 \sigma_2} \right].$$

50 Final Joint Density Function

Thus your final expression is:

$$g(y_1, y_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1 - \rho^2}} \exp \left[-\frac{1}{2(1 - \rho^2)} \left(\frac{(y_1 - \mu_1)^2}{\sigma_1^2} + \frac{(y_2 - \mu_2)^2}{\sigma_2^2} - 2\rho \frac{(y_1 - \mu_1)(y_2 - \mu_2)}{\sigma_1 \sigma_2} \right) \right], \quad -\infty$$

51 Chi-square Distribution

51.1 Chi-square Distribution with One Degree of Freedom

Let

$$X \sim N(0, 1).$$

Define the transformation

$$Y = X^2.$$

We find the pdf of Y .

Since $Y = X^2$, the inverse transformation is

$$x = \pm\sqrt{y}, \quad y > 0.$$

The derivative is

$$\left| \frac{dx}{dy} \right| = \frac{1}{2\sqrt{y}}.$$

The standard normal pdf is

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

Thus,

$$\begin{aligned} f_Y(y) &= f_X(\sqrt{y}) \frac{1}{2\sqrt{y}} + f_X(-\sqrt{y}) \frac{1}{2\sqrt{y}} \\ &= \frac{1}{\sqrt{2\pi}} e^{-y/2} \frac{1}{2\sqrt{y}} + \frac{1}{\sqrt{2\pi}} e^{-y/2} \frac{1}{2\sqrt{y}} \\ &= \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}. \end{aligned}$$

Hence,

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}, \quad y > 0.$$

This distribution is called the **chi-square distribution with 1 degree of freedom**.

51.2 General Chi-square Distribution with k Degrees of Freedom

Let

$$X_1, X_2, \dots, X_k \stackrel{iid}{\sim} N(0, 1).$$

Define

$$Y = \sum_{i=1}^k X_i^2.$$

Then Y is said to follow a **chi-square distribution with k degrees of freedom**. Denoted:

$$Y \sim \chi_k^2.$$

Mean and Variance

$$E(Y) = k, \quad V(Y) = 2k.$$

Moment Generating Function

The mgf of a chi-square variable with k degrees of freedom is

$$M_Y(t) = (1 - 2t)^{-k/2}, \quad t < \frac{1}{2}.$$

52 F Distribution

52.1 Definition

Let

$$X_1 \sim \chi_{k_1}^2, \quad X_2 \sim \chi_{k_2}^2,$$

with X_1 and X_2 independent.

Define the random variable

$$Z = \frac{X_1/k_1}{X_2/k_2}.$$

Then Z has an **F -distribution with (k_1, k_2) degrees of freedom**. We write:

$$Z \sim F(k_1, k_2).$$

52.2 Expectation of F Distribution

The mean exists only when $k_2 > 2$:

$$E(Z) = \frac{k_2}{k_2 - 2}.$$

52.3 Variance of F Distribution

The variance exists only when $k_2 > 4$:

$$V(Z) = \frac{2k_2^2(k_1 + k_2 - 2)}{k_1(k_2 - 2)^2(k_2 - 4)}.$$

52.4 MGF of F Distribution

The moment generating function $M_Z(t)$ **does not exist for any $t \neq 0$** . Thus,

$$M_Z(t) \text{ exists only at } t = 0.$$

53 Student's t -Distribution

53.1 Definition from Normal and Chi-square Distributions

Let

$$X \sim N(0, 1), \quad Y \sim \chi_k^2,$$

and assume X and Y are independent.

Define

$$T = \frac{X}{\sqrt{Y/k}}.$$

Then T follows the **Student's t -distribution with k degrees of freedom**.

$$T \sim t_k.$$

53.2 Interpretation

The numerator is a standard normal variable. The denominator is the square root of an independent chi-square variable scaled by its degrees of freedom. This ratio naturally produces long tails, characteristic of the t -distribution.

54 Conclusion

In these notes we developed the fundamental probability distributions that arise naturally from the normal distribution and its transformations. Beginning with the standard normal variable, we showed how the chi-square, F -distribution, and Student's t -distribution emerge as essential tools for statistical inference.

Summary of Key Results

- The **chi-square distribution** arises as the sum of squares of independent standard normal variables. It forms the basis for variance estimation, goodness-of-fit tests, and hypothesis tests for dispersion.
- The **F -distribution** is constructed as the ratio of two scaled chi-square variables. It plays a central role in comparing variances and in the analysis of variance (ANOVA) framework. Its expectation and variance exist only under specific conditions on the denominator degrees of freedom.
- The **Student's t -distribution** is obtained from the ratio of a standard normal variable and the square root of an independent chi-square variable divided by its degrees of freedom. This distribution is crucial in constructing confidence intervals and hypothesis tests for means when the population variance is unknown.

- Transformations of variables, Jacobian techniques, and moment generating functions provide systematic methods to derive pdfs of transformed random variables. These tools are foundational for understanding the structure and properties of statistical models.

Final Remarks

The distributions studied here form the backbone of classical statistical inference. Many advanced techniques in estimation, hypothesis testing, regression, and machine learning rely on these fundamental results. A solid mastery of the normal, chi-square, F , and t distributions—along with their derivations and properties—provides a strong theoretical foundation for deeper study in probability and statistics.